Are Immigrants More Innovative? Evidence from Entrepreneurs

by

Kyung Min Lee
World Bank and George Mason University

Mee Jung Kim
Sejong University

J. David Brown
U.S. Census Bureau and IZA

John S. Earle
George Mason University

Zhen Liu
George Mason University

CES 23-56 **November 2023**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

To obtain information about the series, see www.census.gov/ces or contact Christopher Goetz, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov. To subscribe to the series, please click here.

Abstract

We evaluate the contributions of immigrant entrepreneurs to innovation in the U.S. using linked survey-administrative data on 199,000 firms with a rich set of innovation measures and other firm and owner characteristics. We find that not only are immigrants more likely than natives to own businesses, but on average their firms display more innovation activities and outcomes. Immigrant-owned firms are particularly more likely to create completely new products, improve previous products, use new processes, and engage in both basic and applied R&D, and their efforts are reflected in substantially higher levels of patents and productivity. Immigrant owners are slightly less likely than natives to imitate products of others and to hire more employees. Delving into potential explanations of the immigrant-native differences, we study other characteristics of entrepreneurs, access to finance, choice of industry, immigrant self-selection, and effects of diversity. We find that the immigrant innovation advantage is robust to controlling for detailed characteristics of firms and owners, it holds in both high-tech and non-high-tech industries and, with the exception of productivity, it tends to be even stronger in firms owned by diverse immigrant-native teams and by diverse immigrants from different countries. The evidence from nearly all measures that immigrants tend to operate more innovative and productive firms, together with the higher share of business ownership by immigrants, implies large contributions to U.S. innovation and growth.

* We thank the National Science Foundation for support (Grants 1262269 and 1719201 to George Mason University) and Thomas Astebro, Bill and Sari Kerr, and participants in presentations at the Society of Labor Economists, American Economic Association, Society of Government Economists, NBER Conference on Immigration, and the General Accounting Office for helpful comments. John Earle was also a Schedule A part-time employee of the Census Bureau while working on this paper. The analysis, thoughts, opinions, and any errors presented here are ours only and do not represent the views of the NSF, U.S. Census Bureau, or the World Bank. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7517031: CBDRB-FY23-CES019-004).

1 Introduction

Do immigrant entrepreneurs create and operate businesses that are more innovative than those owned by native-born Americans? The assumption that immigrant owners contribute disproportionately to U.S. economic growth is implicit in such policies as the EB-5 visa program, which provides a special pathway to U.S. residence for immigrants founding large businesses. Yet immigrants face disadvantages in language, social networks, and other local knowledge that may be important complements to innovation and entrepreneurship. Such factors could especially hinder their ability to innovate in more radical ways and to start up and grow successful businesses, making immigrants more likely to simply imitate local practices and resulting in lower rates of innovation outcomes such as patents and productivity.

On the other hand, immigrant-owned firms might be more innovative if immigrants self-select on personal characteristics such as risk preferences, aspirations, and imaginativeness, which are associated with both innovation and entrepreneurship. Another possibility stems from the stimulating effects of experiencing diverse societies, cultures, languages, and institutions, leading to recombination of ideas that underlie innovation. This mechanism, by contrast with self-selection, raises the intriguing possibility that immigration could increase innovation overall, and therefore potentially raise world economic growth.

Despite the policy relevance and popular interest in immigrants, as yet there has been relatively little systematic evidence on their role as potentially innovative entrepreneurs. A number of studies have examined immigrants as individual inventors, as employees of high-tech firms, and as scientists, engineers, and self-employed, but few have studied immigrant-owned firms. Yet much of consequential innovation takes place within firms, where founder-owners are a key ingredient of innovation output.[1] Previous research on immigrant owners mostly examines firm size and growth, and studies concerned with innovation focus on patenting within small samples of firms in the high-tech sector, as discussed below.

This paper contributes to understanding the innovation impact of immigrant entrepreneurship on the U.S. economy using a much larger, broader, and richer data set than those heretofore available. Much of our analysis exploits data on 199,000 firms from the Annual Survey of Entrepreneurs (ASE), a U.S. Census Bureau survey covering employer businesses based on a random sample of all nonfarm businesses. The ASE questionnaire contains detailed information on the characteristics of the firm's four largest owners, including demographics, human capital, ownership team, use of finance, and industry. Unlike previous data sets con-

[1] As one measure, patents are much more likely to be owned by businesses than individuals. According to U.S. Patent and Trademarke Office Patent Technology Monitoring Team (2014), for instance, 93.1 percent of patents granted in 2014 are owned by corporations, 0.4 percent by governments, and 5.4 percent by individuals. Patterns are similar in other years.

taining owner information, however, and crucially for this paper, the ASE also includes many innovation measures that we study as outcome variables, including reported innovation activities in both products and processes, research and development, trademarks, and patents. We link these data to other data sources on patents, and on revenue and employment, which we use to construct measures of productivity. As an alternative way to measure job creation, we also use Current Population Survey data to analyze shares of immigrants by employment and self-employment status and by the numbers of employees that they hire.

To provide some basic background over a longer time horizon on the contribution of immigrants to entrepreneurship, we compile immigrant shares in population, employment, and self-employment using U.S. Census Bureau data back to 1910.[2] As shown in Figure 1, the immigrant shares for different groups are similar in 1910 and 2021 but show a pronounced U-shape with a trough in 1970. Prior to 1940 the immigrant share in self-employment was lower than the immigrant shares in the adult and employed populations, implying a lower self-employment rate among immigrants during this time period. The different series track each other closely through 2000, rising by 6-7 percentage points between 1970 and 2000. But from 2000 the series diverge with the adult population share rising to about 16 percent, the employment share to 18 percent, and the self-employment share to more than 22 percent. In a reversal of the situation in the early 20th century, the self-employment propensity in recent years is thus much higher for immigrants. The precise difference depends on the comparison: relative to total population, the immigrant self-employment rate is 62 percent higher, relative to adult population it is 37 percent higher, and relative to employed it is 23 percent higher. By any of these measures, however, immigrants are responsible for a disproportionate share of businesses in the 21st century U.S. economy.

In the broad literature on entrepreneurship, the self-employment measure is subject to criticism (e.g., Parker (2004)), as it may reflect outside contracting, casual work, or subsistence or "necessity" activities, and it does not take into account the degree to which the venture is genuinely entrepreneurial in the colloquial sense, creating jobs and innovating with new products or production processes. One approach to come closer to this notion of new business creation is to distinguish incorporated from unincorporated self-employed businesses (e.g., Levine and Rubinstein (2017)). Adopting this perspective, we add information on incorporated and unincorporated self-employment, which is available since 1970, in Figure 2. Before 2000, the immigrant share in incorporated self-employment is higher than for unincorporated, but since then there has been a larger rise in unincorporated than incorpo-

[2] 1910 is the earliest year for which self-employment can be measured. Data are decennial censuses through 2000 and American Community Survey (ACS) data for recent years. Data sources and definitions are described in Section 2.

rated self-employment, reaching 23 and 21 percent, respectively. Both the incorporated and unincorporated shares of immigrants exceed their employment share throughout this period, but immigrants are less likely to incorporate their businesses. These calculations include all workers regardless of their hours, but working longer hours might also be taken as a proxy for more committed entrepreneurship. Figure 2 also contains the share of immigrants in full-time self-employed. This variable is also similar to the immigrant share in employment and self-employment until about 2000, after which it rises more rapidly. By 2021, it reaches 23 percent. By this measure, immigrants appear to be more committed entrepreneurs.

Our analysis explores these household data further below, but most of the paper focuses on firm-level data for which we have direct measures of innovation activities and outcomes. Previous studies of innovation and immigrant ownership rely on small sample surveys of businesses in particular industries, regions, or owner ethnicities, the innovation measures are usually only patents and R&D, and they focus almost exclusively on the high-tech sector. For instance, Hart and Acs (2011) examine these innovation measures using a survey of 1,300 "high impact" high-tech companies, finding little difference between firms with and without immigrant founders.[3] An exception to the small survey samples is Brown et al. (2020), who also use the ASE, but focus only on survey responses in high-tech firms. We build on that paper in several ways. First, in order to estimate immigrant-native differences in innovation behavior more broadly, we examine the whole economy and compare high-tech and low-tech sectors. Patenting and R&D are more common in the high-tech sector, but a significant amount of innovation activity occurs in the low-tech sector as well, and the low-tech sector makes up most of the economy. We include data from not only survey responses on patent holding, but also registered U.S. Patent and Trademark Office (USPTO) patent data. We examine firm employment and productivity as additional outcome variables, both at the mean and at the top of the distribution, to see whether immigrant-owned firms are more likely to be industry leaders along these dimensions. Finally, we examine the effects of ownership team diversity on all these firm innovation and outcome measures.

A few other studies conduct firm-level research on immigrant entrepreneurs using large, nationally representative samples. Fairlie and Lofstrom (2015) reviews such studies from the 1990s and early 2000s. More recently, Kerr (2017*a*) uses data from the Longitudinal Employer-Household Dynamics (LEHD) to identify immigrant-owned firms and examine various measures of firm performance in the Primary Metropolitan Statistical Areas of eleven U.S. states. Focusing on new start-ups, he defines immigrant-owned firms as those with at least one immigrant among the employees in the top three earners for the first three years

[3]Other related studies include Saxenian (2002), Wadhwa et al. (2007*a*), Wadhwa et al. (2007*b*), Hunt (2011), Kerr (2008), and Kerr and Kerr (2016).

after entry. With this definition he finds rates of immigrant ownership increasing to 27 percent of employers in 2008, the last year in their data. Using the 2007 Survey of Business Owners (SBO) for new entrants that year, Brown et al. (2019) report that 15 percent of firms are wholly owned by immigrants and another 4.3 percent are owned by mixed immigrant-native teams. Kerr and Kerr (2020) use the SBO for 2007 and 2012 and report that the share of firms less than five years old with an immigrant owner is 24 and 26 percent, respectively. Immigrant-owned firms have fewer employees, pay lower wages, and provide fewer fringe benefits, but their receipts per employee are higher.

Concerning individual inventors, studies document higher rates of STEM workforce participation, patents, publication citations, and Nobel Prize winners among immigrants.[4] Akcigit, Grigsby and Nicholas (2017) provide evidence from U.S. patent data linked to decennial censuses between 1880-1940 that immigrants have more patents and citations than natives. Bernstein et al. (2022) link patent data to Social Security numbers and birth year to approximate immigrant status and find that immigrants represent 16 percent of all inventors in the U.S., producing 23 percent of total innovation output, measured by number of patents, citations, and the economic value of the patents.[5]

Yet much innovation takes place within firms, and our study relates to research on firm-level Research and Development (R&D), patenting, and other aspects of innovation. As is widely recognized, however, R&D and patents both have limitations as measures of innovation, much of which takes place without formal R&D or patenting. Some surveys, including the Community Innovation Surveys (CIS) in Europe and the Business Research and Development and Innovation Survey (BRDIS) in the U.S., attempt to fill this gap with qualitative questions on product and process innovations (Mairesse and Mohnen, 2010). These surveys have documented the incidence of such activities and demonstrated their correlation with productivity (e.g., Griffith et al. (2006) Parisi, Schiantarelli and Sembenelli (2006), Hall (2011). But the data in these studies are usually based on small samples (for example, only 5,000 firms receive the full questionnaire for the BRDIS) that are non-randomly selected to focus on firms with known R&D activity. For our purposes, a crucial gap in these surveys is that they contain no information on the firm's founders or owners.

Finally, our paper is also related to studies of immigration and labor markets, including effects on native worker wages (e.g., Borjas (2003), Card (1990), Card (2001), and Ottaviano

[4]See for instance Stephan and Levin (2001), Hunt and Gauthier-Loiselle (2010), Kerr and Lincoln (2010), Kahn and MacGarvie (2016), Kerr (2017*b*), and papers in the special issue in this journal (Breschi et al., 2020).

[5]Using data on patenting by country and product class, Bahar, Choudhury and Rapoport (2020) report higher patenting in products associated with increased migration from countries specialized in those products. Using Australian regional data, Crown, Faggian and Corcoran (2020) find that a skilled immigration visa program is positively associated with the number of patent applications.

and Peri (2012)) and the extent of immigrant-native convergence in wages, or "assimilation" (e.g., Borjas (1985), Chiswick, Lee and Miller (2005), Chiswick (2009)). Issues from these literatures concerning the degree of substitutability between immigrants and natives and the potential disadvantages faced by immigrants arise in the context of our analysis of immigrant entrepreneurs. If immigrant-owned firms tend to be more innovative and productive than native-owned, this would suggest that immigrants are not perfect substitutes for native entrepreneurs. It would also imply that whatever disadvantages immigrants face in assimilation, they are able to overcome them through their innovation activities.

The next section contains a description of our data, including the sources and definitions of the measures of innovation activities and outcomes. The third section describes our empirical methods, and the fourth contains results, divided into subsections for unconditional differences in innovation, regression results controlling for alternative sets of covariates, innovation in high-tech vs. low-tech firms, and an analysis of co-ownership diversity and innovation. A brief section concludes the paper.

2 Data

2.1 Data Sources and Samples

Our analysis of the role of immigrant entrepreneurs in innovation combines data from several sources, some of which have never before been studied for this purpose. The database provides a much broader set of innovation measures and enables us to examine some alternative hypotheses for differences between immigrant- and native-owned firms.

The principal source for most variables is the 2014 Annual Survey of Entrepreneurs (ASE) from the U.S. Census Bureau. The ASE collected information on businesses in the nonfarm sector with at least one paid employee and annual receipts of $1,000 or more. The ASE sample is stratified by the 50 most populous Metropolitan Statistical Areas (MSAs), state, and the firm's number of years in business. Firms are randomly selected based on volume of sales, payroll, or number of paid employees, with the exception that large firms are selected with certainty in each stratum. The initial 2014 ASE sample size is about 290,000 employer firms with a response rate of 74 percent.

The unusual strength of the 2014 ASE, particularly relevant for this study, is its large number of innovation measures. These measure different types of product and process innovation, R&D activities, and intellectual property (copyrights, trademarks, and patents), which serve as some of the dependent variables in our analysis.[6] We describe these variables

[6] Lee et al. (2022) also examine innovation measures from the ASE, but with a focus on African-American

in detail below.

The ASE also collects characteristics on up to four owners with the largest shares in the business, among those with at least 10 percent. Our main variable of interest is an indicator for whether the business is owned by an immigrant, which is defined by the ASE as a noncitizen at birth. The information on multiple owners allows us to study entrepreneurial teams, including two types of diverse teams. The first type of diversity, "diverse native-immigrant," is defined as at least one immigrant and at least one native owner. The second type, "within-immigrant diversity," relies on the detailed race and ethnicity categories in the ASE that enable us to identify region and in most cases the specific country of origin and to identify cases where at least two immigrant owners have different origins. We construct five ownership categories: (1) "Any Immigrant" for firms with at least one immigrant owner; (2) "Only Immigrant" for firms with only immigrant owners, (3) "Diverse Native-Immigrant" when firms have mixed immigrant and native owners, (4) "Diverse Within-Immigrant" for firms with immigrants from different racial or ethnic backgrounds, and (5) "Only Native" for firms with only native owners. Because the samples for (3) and (4) are much smaller than for the other categories, we report results for them in a separate subsection.

We have linked the ASE with the Census Bureau Patent Crosswalk Data (CBPCD), the Business Register (BR), and Longitudinal Business Database (LBD). The CBPCD contains administrative records on the number of patents granted to each firm by year (Dreisigmeyer et al., 2018; Graham et al., 2018). The BR provides number of employees and revenue for the employment and productivity analysis. The LBD contains firm age (based on the entry year of the firm's oldest establishment), an essential control variable given the strong association between firm age and many dimensions of innovation. Together, these data provide additional measures of innovation performance.

We construct two firm-level samples for analysis. One is the regression sample based solely on ASE data. The second involves the linked ASE-LBD-BR-CBPCD regression sample. The first sample is used for the analysis of innovation activities, and the second sample is used to examine innovative outcomes. We restrict the samples to firms with non-missing values across both dependent and independent variables. The final ASE regression sample includes 199,000 firms, and the final sample of the linked ASE-BR-CBPCD data contains 135,000 firms. We also construct new weights addressing missing observations by multiplying the ASE sampling weight by the inverse probability of an ASE firm being in the regression sample.

For our analysis with household data, our sources are the decennial census, American Community Survey (ACS) and Current Population Survey (CPS). While the decennial census

owned firms.

is conducted every ten years, the ACS is conducted every year since 2001. The earliest information on immigrant status is in the 1910 Census. We calculate immigrant shares in the U.S. population, adult population, employment, and self-employment from that year until 2021. Starting in 1970, further information on type of self-employment, including incorporation and full-time status, becomes available. The CPS, a nationally representative survey of U.S. households, provides detailed demographics and labor force information. We use immigrant shares in population, employment, and self-employment in 1994, the first year that the CPS included a regular question on immigration status. We also exploit new data on the number of employees reported from 2014 to 2019 by self-employed individuals in questions added to the Outgoing Rotation Groups (ORG) questionnaire since 2014.

In accordance with the definition of immigrants in the ASE, our analyses for the decennial census and the ACS consider naturalized citizens and noncitizens as immigrants, and those born in the U.S., U.S. outlying areas, or of American parents abroad as natives. Since the 1910 and 1960 Census do not have complete citizenship information, we use only birthplace to define those born outside the U.S. as immigrants for these two years.[7] As a result, 1910 and 1960 immigrant shares could be overstated, as those born abroad of American parents cannot be distinguished from immigrants without the citizenship status variable.

For the employed or self-employed sample in the CPS and ACS analysis, we exclude those who have jobs in armed forces or public sectors because noncitizens are not permitted to work in those sectors. We also exclude agriculture as the self-employment share is disproportionately high in this sector. This sample restriction makes the CPS and ACS analysis consistent with ASE-LBD analysis that is limited to the non-farm private sector.

2.2 Variables and Descriptive statistics

Data limitations in previous research on innovation have led to a nearly exclusive focus on intellectual property (patents) and, to a lesser extent, R&D. But these measures have clear drawbacks, particularly outside the context of the very small set of industries within which firms compete to patent and have the resources for formal R&D departments. We also study patents and R&D but take a broader approach to innovation, following the OECD/Eurostat definition: "the implementation of a new or significantly improved product (good or service), or process, a new marketing method, or a new organizational method in business practices, workplace organization or external relations" (OECD/Eurostat, 2018). We organize the various measures we have assembled from the sources described above into two categories: innovation activities, which include six types of product innovation, six types of process

[7]For the 1970 Census, we use Form 1 for the citizenship question. The appendix contains figures with the definition of immigrant in all years as foreign-born regardless of citizenship status at birth.

innovation, and two types of R&D; and innovation outcomes, which include intellectual property - trademarks and patents, employment, productivity, and industry leadership (top 5% of the distribution).

Detailed definitions are provided in Table 1a for the innovation activity measures and in Table 1b for the innovation outcomes. The innovation count variable is a continuous variable that sums up the number of product and process innovation activities, among the 12 listed, that the firm reports carrying out in 2012-2014. Any innovation is a dummy variable for whether the firm reports any of the 12 activities, while any product and any process are dummies for carrying out any of the six product and six process innovation activities, respectively. The remaining product and process innovation, R&D, and intellectual property variables are all dummy variables. We have constructed the R&D variables by grouping detailed survey questions into "basic" and "applied" types, as shown in the table. Employment is the logged number of employees and productivity is calculated as the firm's log of (revenue divided by number of employees) minus the 4-digit NAICS industry mean of that variable in 2014. Employment top 5% and productivity top 5% are dummy variables for whether the firm was above the 95th percentile of employment and productivity in 2014, respectively. We provide means for all these variables by ownership type at the beginning of the results section.

Descriptive statistics for firm and owner characteristics by ownership type are shown in Table 2. About 19% of firms are immigrant-owned (Any Immigrant).[8] Immigrant-owned firms and especially those owned only by immigrants tend to be younger than native firms, and a higher share of only immigrant-owned firms have one owner than only native-owned firms (67 percent compared to 59 percent). Most diverse firms, either diverse native-immigrant or within-immigrant, have two owners (65 percent and 58 percent, respectively), and the remainder have three or more. More than half of all firms are owned by only male owners. Diverse immigrant owner firms are more likely to have female owners compared to only native or only immigrant firms, possibly just because the former must have at least two owners. Mixed-gender unrelated ownership is equally uncommon for only immigrant and only native firms (3.7 percent for both), but diverse native-immigrant firms are more likely to be owned by family members (48.6 percent). Immigrant firm owners tend to be younger, with only 36 percent above 54 years old in only immigrant firms compared with 50 percent of only native firms. All types of immigrant owners, and particularly those in diverse within immigrant firms, are more likely to have advanced degrees and prior business experience.[9]

[8] Kerr and Kerr (2020) report that about 25% of firms under age 5 in the 2007 and 2012 SBO are immigrant-owned.

[9] Baum et al. (2022) suggest that knowledge transfer via board members' prior experience and networks increases the probability that a new firm is innovative, measured by number of patent applications.

Not surprisingly, owners of only native firms are more likely to be U.S. veterans.

Table 3 contains further information on firm characteristics, showing differences in the amount and sources of start-up capital and in industry by ownership type. Immigrant-owned firms tend to have more start-up capital compared to only native firms. This suggests that financial constraints may not be a bigger problem for immigrants, and it could reflect the different plans and aspirations the entrepreneur has when starting the firm. Both types of diverse immigrant firms have much larger amounts of start-up finance, with 31 percent of the diverse within-immigrants having at least 100k, compared to 18 percent of only native firms, although this could reflect differences in the number of owners. Regarding sources of start-up finance, immigrant firms are more likely to use personal savings (75.4 percent) compared to only native firms (67.3 percent). On the other hand, only native firms use more bank loans compared to immigrant firms (19 vs. 14 percent). Diverse within-immigrant firms are more likely to have outside investors compared to other immigrant firms and only native firms, which again could reflect differences in the number of owners across these groups.

In terms of industry, immigrant-owned firms are much more concentrated in accommodation and food industry, retail trade, or health, whereas only native firms are more often in the professional services and construction sectors. Lastly, immigrant firms are more likely to be in the high-tech sector (here defined on the basis of narrower industries with either a large STEM workforce or high R&D) compared to only native firms (6.4 percent vs. 4.9 percent).[10] Both types of diverse ownership are more likely to be in high-tech, but the highest rate by far is for diverse within-immigrant firms, where the share in high-tech is 13 percent.

3 Methods

To understand the impact of immigrant ownership on innovation, we first quantify the raw gap through descriptive statistics. Next, focusing on differences in innovation between firms with any immigrant owner compared with only native owners, we estimate the gap and how it changes in regression specifications with various sets of controls. Results are shown for three different regression specifications. The first "base" specification includes firm age, ownership team size and family relationships, detailed characteristics of the entrepreneur

[10]The high-tech sector includes these 4-digit NAICS industries: Oil and Gas Extraction (2111), Pharmaceutical and Medicine Manufacturing (3254), Computer and Peripheral Equipment Manufacturing (3341), Communication Equipment Manufacturing (3342), Semiconductor and Other Electronic Component Manufacturing (3344), Navigational, Measuring, Electromedical, and Control Instruments Manufacturing (3345), Aerospace Project and Parts Manufacturing (3364), Software Publishers (5112), Wired and Wireless Telecommunications (5171), Other Telecommunications (5179), Data Processing, Web Hosting, and Related Services (5182), Other Information Services (5191), Architectural, Engineering and Related Services (5413), Computer System Design and Related Services (5415), and Scientific Research and Development Services (5417).

(age, gender, education, prior business experience, etc.), and the indicator for having at least one immigrant owner (any immigrant) on the right-hand side.[11] The second specification adds a set of variables measuring the amount of start-up finance, and the third adds 4-digit industry dummies. Both the amount of finance and the industry in which the firm operates are choice variables to some extent, and thus may be endogenous in these equations. But the results help illuminate the degree to which these factors condition the immigrant-native differences.

We estimate the impact of immigrant ownership on innovation behavior using linear regressions. The general specification for differences in immigrant- versus native-owned firms is the following:

$$Y_j = \beta_0 + \beta_1 AnyImmig_j + \boldsymbol{X_j}\boldsymbol{\gamma} + \boldsymbol{K_j}\boldsymbol{\alpha_K} + \boldsymbol{S_j}\boldsymbol{\alpha_S} + \epsilon_j \quad (1)$$

where Y_j are measures of innovation activities (product and process), intellectual property, employment, or productivity for a firm j. $AnyImmig_j$ is an indicator for a firm with any immigrant owner, which is our main variable of interest. $\boldsymbol{X_j}$ is a vector of characteristics of firm and its owner(s) j, including firm age, owner demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). $\boldsymbol{K_j}$ is a set of detailed categories of start-up finance amounts. $\boldsymbol{S_j}$ is a set of 4-digit NAICS industry dummies.

In the base specification, we control only for pre-determined firm, owner, and ownership team characteristics, not including finance ($\boldsymbol{K_j}$) and sectors ($\boldsymbol{S_j}$). In the second specification, we add start-up finance controls to see how they change the immigrant gap in innovation. As also found by Kerr and Kerr (2018), we showed in the Data Section that immigrants use higher levels of finance when starting up their businesses, which suggests that financial market discrimination is not a systematic problem for immigrants. It may instead reflect differences in demand for finance associated with the ideas and plans for the business. The finance variable may thus be partly endogenous, which is why we exclude it from the base specification. Another potential control, also endogenous to planned outcomes, is the choice of industry in which the business operates. But firms are obviously quite heterogeneous across industries, so within-industry comparisons are of interest. Therefore, in the third specification, we add sector controls to address this heterogeneity.

In addition to the ASE, we use data from the CPS ORG to examine differences in entrepreneurial propensity with the following regression equation:

[11] We have estimated several other versions of these equations, for instance excluding human capital variables, but these made little difference to the results, so we show only a single "base" specification including all these variables.

$$Y_i = \theta_0 + \theta_1 Immig_i + \boldsymbol{X_i\eta} + \boldsymbol{S_i\mu_S} + \epsilon_j \tag{2}$$

where Y_i are measures of an individual's entrepreneurial status in alternative specifications: self-employed, incorporated self-employed, employer self-employed, and employer with more than a certain number of employees (e.g., 5+, 10+, 20+, and 50+). $Immig_i$ is an indicator for an individual who is not born a citizen (including both naturalized citizens and noncitizens). $\boldsymbol{X_i}$ is a vector of characteristics of an individual i, including demographics (female, age, and married) and human capital variables (education and veteran). $\boldsymbol{S_i}$ is the set of 2-digit industry dummies.

To understand the differential impacts of ownership diversity, we replace the any immigrant owner variable in equation (1) with three ownership indicators: only immigrant, immigrant-native diverse, and within-immigrant diverse. Then we estimate linear regressions with these three variables using the following extension of the above equation:

$$Y_j = \delta_0 + \delta_1 OnlyImm_j + \delta_2 DivNatImm_j + \delta_3 DivWithinImm_j + \boldsymbol{X_j\gamma} + \boldsymbol{K_j\alpha_K} + \boldsymbol{S_j\alpha_S} + \epsilon_j \tag{3}$$

where $OnlyImm_j$ is an indicator for a firm with only immigrant owners, $DivWithinImm_j$ is a dummy variable for a firm with differences in ethnicity or race among immigrant owners, and $DivNatImm_j$ is an indicator for a firm with both immigrant and native owners. As in the previous specifications, we start with the base specification containing only predetermined owner and firm characteristics (including number of owners) and then successively add controls for finance and industry in additional specifications.

4 Results

This section describes our results for differences in innovation activities and outcomes by ownership type in four subsections: unconditional differences, regression results for any immigrant owner, results for high-tech and low-tech firms separately, and results for native-immigrant and within-immigrant diversity.

4.1 Unconditional Immigrant-Native Differences in Innovation

Table 4a shows mean differences by ownership type in innovation activities. Compared to only native firms, immigrant firms are with one exception more likely to conduct any and all types of the measured innovation activities, including product and process innovation and R&D. The exception is new product for the firm (but not for the market), where only

native ownership has a small advantage over any immigrant ownership (15.4 versus 14.5 percent), possibly reflecting a lower tendency for immigrants to imitate native firms. The incidence of all types of innovation activities is much higher for diverse native-immigrant and within-immigrant ownership. For example, while the rate at which firms introduce a completely new product is about 53 percent higher for firms with any immigrant ownership compared with only native, it is 85 percent higher with diverse native-immigrant ownership and 120 percent higher with diverse within-immigrant ownership. The differences are even more pronounced for both basic and applied R&D. Note, however, that these unconditional differences do not control for basic variables like firm age and number of owners, results for which are provided in the next subsection.

Turning to measures of innovation outcomes, Table 4b shows unconditional mean differences for intellectual property, employment, and productivity. The pattern observed for innovation activities holds still more strongly for patenting behavior. For instance, firms with any immigrant owner are 67 percent more likely to receive at least one new patent in both the 2012-14 and 2015-2017 periods but three to four times more likely under diverse native-immigrant ownership and about six to eight times more likely with diverse within immigrant ownership. The rate at which firms receive three or more is about 10 times higher with diverse within-immigrant than with only native ownership in 2012-14 (for the later period, the sample size is too small to permit disclosure of the exact result).

Concerning employment, immigrant firms have a slightly lower number of employees on average and are less likely to be in the top 5 percent of the employment distribution. Both types of diversity are associated with larger employment, but this could be associated with the number of owners. On the other hand, immigrant firms' average productivity is higher, and they are about 50 percent more likely to be in the top 5 percent of the industry-specific productivity distribution than only native firms.

The employment results may be somewhat surprising, so we consider an alternative source for immigrant-native firm employment analysis: the CPS.[12] Unlike the ASE, it permits an analysis of relative propensities of immigrants to be self-employed, and for recent years it contains information on number of employees.

The first bars in Figure 3 shows similar results to those displayed above using decennial censuses and the ACS, including the rise in immigration since 1994 (the first year that immigration questions were included on the CPS), the substantially higher rate of self-employment for immigrants, and the lower rate of incorporation among the immigrant self-

[12]Consistent with the definition of immigrant in the ASE, our analysis for the CPS defines people not born a citizen (naturalized citizens and noncitizens) as immigrants, and those born in the U.S., U.S. outlying areas, or abroad of American parents as natives.

employed. The share of immigrants among employers (self-employed who have employees) is 18.9 percent on average over 2014-19, which is higher than their share in employment (18.1) but lower than their share of self-employed (20.5). Studies of employer probabilities (e.g., Fairlie and Lofstrom, 2015) usually take existing businesses (including nonemployers) as the base, which here would imply a comparison of the employer share and the self-employment share. But if the question is about immigrants' contributions, then the share in employment (18.1 percent) or population (14 percent overall and 16 percent among those age 16 and older) may be more appropriate. The relevant comparison group is an issue to which we will return when analyzing firm data and in drawing conclusions on the role of immigrants in U.S. innovation.

To address the question of how much job creation takes place in firms owned by self-employed immigrants, the bars on the right side of Figure 3 show the immigrant shares in size groups based on various thresholds. The immigrant share in firms with at least 5 employees (16.5 percent) is similar to their total employment share, but the immigrant share declines with size, so that among firms with 50 or more employees, the share is only 10 percent. The mean number of employees for immigrant self-employed is 6.7, compared with 8.9 for native self-employed. The difference is likely understated because the publicly-available variable is top-coded at 75, which is at or below the 99th percentile of the native distribution, while 72 is the 99th percentile for immigrants. These CPS data imply that immigrant-owned firms tend to be smaller, undercutting the conventional wisdom that immigrants create a disproportionate share of the largest firms. We hasten to add that the CPS is unlikely to capture larger businesses accurately, because of the small CPS sample size, the tiny share of such firms, and the presence of multiple owners, many of whom do not report themselves as self-employed operators of the businesses. However, these results are consistent with our firm-level analysis, which also shows smaller numbers of employees in firms with only immigrant owners.

Of course, all of these comparisons of means do not control for other factors that may affect innovation activities and outcomes. To consider such factors, we next turn to regression results.

4.2 Regression-Adjusted Differences for Any Immigrant Owner

We organize the regression results in tables by showing the unconditional mean for only native-owned firms in the first column, to provide a baseline for interpreting the magnitudes. The other columns contain coefficients (and standard errors) for any immigrant estimated for the three regression specifications explained in the Methods Section: base specification with

firm size, number of owners, and owner demographic and human capital characteristics; the same plus finance measures; and the same plus four-digit industry.[13] Table 5a shows results for any immigrant in regressions where the innovation activities are the dependent variables. In the base specifications, firms with any immigrant owner have statistically significantly higher propensities to engage in all the innovation activities with the exception of developing a product that is new to the firm, but not the market. Some of the effect magnitudes are quite large. For completely new product, the effect is 2.1 percentage points relative to a native-only mean of 4.7 percent (implying a 45 percent increase), and for the new use product innovation, the effect is 6.3 percentage points relative to a native-only mean of 7.9 (80 percent). Among process innovation activities, the effects are especially large and robust for innovations to reduce production cost and improve delivery, where the impacts of any immigrants are each about 20-25 percent. For basic and applied R&D, the effects are 44 and 39 percent, respectively. The effects are smaller for other measures. The effect on the number of innovations is 0.37 of an innovation, or an increase of 18 percent, and the effect on the incidence of any innovation is 6.5 percent. Adding finance and industry controls reduces the size of the estimated effects. But they remain substantial and statistically significant for all measures except upgraded technique.

The estimated effects of an immigrant owner on patenting are large and statistically significant, as shown in Table 5b. The effect is 35 percent on having a patent using the ASE question, 61 percent for receiving a patent in 2012-2014 in the CBPCD, and 70 percent for receiving a patent in 2015-2017 in the CBPCD. The results are nearly identical when including finance and industry controls. For copyrights or trademarks, the immigrant effect is small and only marginally significant, and it is insignificant when including finance and negative when including industry controls.

The immigrant effects on average employment and the propensity to be in the top 5 percent of the employment distribution are negative. The effect magnitudes increase when including finance and industry controls, suggesting that immigrants tend to own firms that have more start-up capital and are in industries with larger average size. By contrast, the average and top 5 percent productivity effects are positive, large, and little affected by finance and industry controls. The propensity for an immigrant-owned firm to be in the top 5 percent of the productivity distribution is 40 percent (1.9/4.6) larger than for an only native-owned firm.

Providing further information about job creation by immigrant owners, Table 6 shows

[13] We have estimated the base equations with various combinations of the demographic and human capital characteristics, and the results are very similar to the first specification shown here, so we omit those other specifications here.

regression results based on the same CPS ORG data used for Figure 3, but with samples restricted to employed individuals for the self-employment analysis and to self-employed for the analysis of employers and their numbers of employees. The base specification has no controls. Demographic and human capital variables are included in the second specification, and the third adds 2-digit industry effects. Among employed individuals, immigrants are more likely to be self-employed in the base specification. When controlling for industry, the effect of being an immigrant on the propensity to be self-employed turns negative and statistically significant, suggesting that immigrants are more prevalent in industries where self-employment rates are high. Among the self-employed, immigrants are less likely to be employers, and that effect is increasing in absolute value in firm size. The effect is about -60 percent in the 50 or more employee firm size class. The coefficients for the propensity to be an employer and to be an employer with a higher number of employees are even more negative when controlling for industry. These patterns corroborate the employment results in Table 5b using the ASE.

4.3 High-Tech and Low-Tech Sectors

Much of the research on innovation focuses on the high-tech sector, perhaps because the most common measure of innovation is patents, which are much more common in that sector. Yet most of the economy (95 percent of the firms in our sample) is not high-tech, and innovation in what we label "low-tech" also affects welfare and growth. With this motivation, Tables 7a and 7b contain results for similar regressions to Tables 5a and 5b, separately for high- and low-tech sectors. To save space, we show only the base specification in the text. Results from specifications including finance and industry controls are qualitatively similar (except as noted below) and can be found in the appendix. We consider a limited set of innovation activity variables because of the small sample size in the high-tech sector together with the small means for some dependent variables.

In general, the immigrant advantage in the variables representing product and process innovation activities is much larger in the low-tech sector. As before, effects are attenuated somewhat when including finance and industry controls. The other product innovation and process innovation effects in the high-tech sector become insignificant. Both types of R&D show large immigrant effects in both sectors, however.

The immigrant effects on patents are positive in both the high- and low-tech sectors, but many of the coefficients in low-tech are imprecisely estimated. The high-tech sector effect size is generally larger for these variables, implying for several of them a doubling of patenting in firms with any immigrant ownership relative to only native ownership, results

that are robust to including finance and industry controls.

It is notable that the negative immigrant employment effect we reported in 5b is concentrated in the low-tech sector. The immigrant coefficients in the high-tech sector regressions for log employment and the probability of being in the top 5 percent of the employment distribution are statistically insignificant. Positive immigrant productivity effects are found in both sectors, implying about 8 percent higher in high-tech and 10 percent higher in low-tech, as well as much higher probabilities of being industry productivity leaders (in the top 5 percent of the industry-specific productivity distribution).[14]

4.4 Ownership Diversity

In this final subsection we study the effects of ownership team diversity on innovation. Immigrant owners could have a high propensity to innovate because more innovative individuals are more likely to immigrate to the U.S. (selection) or because they benefit from different perspectives (diversity).[15] Our regressions control for many characteristics that may be associated with the propensity to innovate, including education and prior business ownership, which should reduce the role of selection in the remaining immigrant-native gap in innovation. Moreover, some factors that we control for may be affected by the propensity to innovate, such as finance. But selection and diversity may also be driven by unobservables, so they are difficult to disentangle empirically.

However, if the diversity mechanism is important in generating innovation, then it would stand to reason that within-firm ownership diversity could increase the variety of perspectives in the firm. If diverse perspectives are a mechanism through which immigrant owners affect innovation, then one would expect diverse ownership teams to be more innovative. We test this diversity of perspectives hypothesis by replacing the any immigrant dummy with dummies for ownership teams with only immigrant owners, teams with both immigrant and native owners (native-immigrant diversity), and those with immigrant owners whose countries of origin, based on the very detailed race and ethnicity questions in the ASE, differ from one another (within-immigrant diversity). Only immigrant and diverse native-immigrant are mutually exclusive. However, diverse within-immigrant is nested in either only immigrant (80 percent of the cases) or in diverse native-immigrant (20 percent), so that

[14] Although their samples, time period, and measurement differ from ours, Kerr and Kerr (2020) report similar employment results, with estimated differences of immigrant versus native ownership of -9.6% in low-tech and -2.4% in high-tech, but more modest immigrant effects on productivity.

[15] In related work, Parrotta, Pozzoli and Pytlikova (2014) and Laursen et al. (2020) find that diversity within a firm's workforce raises patenting. Bercovitz and Feldman (2011) find that academic teams comprised of members from diverse institutions with different experiences are more likely to generate patents. Ferrucci and Lissoni (2019) report higher patent quality with inventor team diversity. Posch, Schulz and Henrich (2023) find that surname diversity within a U.S. county raises patents at the county level.

the diverse within-immigrant coefficient is additive with respect to those categories.

Tables 8a and 8b show the base specification, and specifications including finance and industry controls are in the appendix. For most innovation activities, the only immigrant and diverse native-immigrant coefficients are positive, statistically significant, and of similar magnitude, implying that native-immigrant diversity does not add much to the immigrant advantage. In contrast, most within-immigrant diversity coefficients are large, showing additional gains to the other two categories when diverse immigrants are owners. They are often imprecisely estimated, though. The magnitudes often imply a much higher rate - doubling or greater - in the immigrant advantage when diverse immigrants are owners. Both kinds of immigrant diversity, but especially within-immigrant, are strongly positively associated with basic and applied R&D.

With respect to innovation outcomes, immigrant diversity of both types is positively associated with owning patents, copyrights, and trademarks. The rate of patenting is much higher in firms with diverse within-immigrant ownership, for some variables more than 10 times that for only native-owned firms. However, firms owned by diverse teams are not estimated to be more productive and may even be less so than homogeneous immigrant-owned firms. Given all the positive diversity results for innovation activities, R&D, and patenting, it seems inconsistent that diversity is not associated with higher productivity. Perhaps these firms tend to be start-ups aiming less at maximizing current revenue and more at being acquired in the future, which our productivity measure cannot capture.

5 Conclusion

Most previous research on innovation by immigrants has focused on individual inventors and scientists and used patents as the principal measure of innovation. But there are several advantages to examining the innovation contributions of immigrants at the firm level and to examining innovation measures besides patents. A large share of innovation takes place within firms. Moreover, much innovation is never patented. Patents are best at capturing innovation in high-tech sectors but are rare in most of the economy. At the firm level, there are many additional measures available through innovation surveys, like those we analyze in this paper, which provide a much richer picture of the dimensions and types of innovation that immigrants contribute.

Our research finds strong evidence that firms owned by immigrant entrepreneurs tend to be substantially more innovative than firms owned by natives. The immigrant advantage holds for both product and process innovations, for both basic and applied R&D, for measurable outcomes of these activities such as patents and productivity, and in both the high-tech

and low-tech sectors. Immigrant entrepreneurs are especially active innovators along such dimensions as creating completely new products, finding new uses for products, changing processes to reduce costs, and engaging in basic R&D, and they are disproportionately located on the right tail of outcome distributions, as top performers in terms of the number of patents and the probability of having productivity in the top five percent of their industry distribution. The exceptions are few: the one type out of twelve product and process innovation activities in which immigrants do not dominate is adopting (imitating) a product of another firm, and the one outcome lower among immigrant-owned firms is employment (a result concentrated in the low-tech sector).

Investigating diverse ownership of two types, native-immigrant and within-immigrant diversity, we find similar positive effects of immigrant entrepreneurs on product and process innovation activities, but amplified effects on R&D and patenting. Patenting is especially higher for the diverse within-immigrant owner teams, but productivity effects for this group are weaker and statistically insignificant.

Among the interesting questions for future research sparked by these results is the source of the immigrant advantage. We can distinguish three types of explanations based on differences in constraints, preferences, and skills. Kim (2019) has argued that immigrants become entrepreneurs because of greater difficulties finding jobs as wage-and-salary employees. Or perhaps immigrants have stronger preferences to be business owners or have a higher tolerance for risk. These explanations could account for higher self-employment rates, but they do not explain the much higher productivity and innovation at immigrant-owned firms.

Thus, the more likely explanation for the mean differences in innovation seems to be related to skills. The regressions we have presented (specifications from "base") control for observables including education, former business ownership, and military experience. The relevant skills may be unobserved, but an interesting question is whether they reflect self-selection of immigrants with higher innovation skills than the general population or, alternatively but not mutually exclusively, that there is a causal effect of immigration that increases innovation. The former could happen because immigration is itself an innovative act. The latter could occur if immigrants can more easily spy market opportunities, gaps that could be filled, as a result of their experience of different cultures and institutions. Yet another possibility is that the ability distribution of immigrants has a fatter right tail than for natives, and that entrepreneurs are drawn from that right tail, leading to more immigrants with higher skills entering entrepreneurship (Åstebro, Chen and Thompson (2011); Kahn, La Mattina and J MacGarvie (2017)).

Our conclusion that immigrants are more likely to carry out innovation activities, obtain patents, and raise productivity more than natives suggests that immigrants have positive

externalities for U.S. economic growth and welfare. Of course, there may be displacement as well, and Fairlie and Meyer (2003) report evidence of crowding out of native self-employed, although they also find positive effects on native self-employment earnings. As in the debate over the impact of immigration on native worker wages, the extent of displacement depends on the similarity of immigrant and native skills and the degree to which they are substitutes or complements. Our findings concerning productivity and innovation suggest that immigrant skills are at least somewhat different, and that immigrant entrepreneurs make substantial contributions to U.S. economic growth.

References

Akcigit, Ufuk, John Grigsby, and Tom Nicholas. 2017. "Immigration and the rise of American ingenuity." *American Economic Review*, 107(5): 327–331.

Åstebro, Thomas, Jing Chen, and Peter Thompson. 2011. "Stars and misfits: Self-employment and labor market frictions." *Management Science*, 57(11): 1999–2017.

Bahar, Dany, Prithwiraj Choudhury, and Hillel Rapoport. 2020. "Migrant inventors and the technological advantage of nations." *Research Policy*, 49(9): 103947.

Baum, Christopher F, Hans Lööf, Andreas Stephan, and Ingrid Viklund-Ros. 2022. "Innovation by start-up firms: The role of the board of directors for knowledge spillovers." *Research Policy*, 51(1): 104375.

Bercovitz, Janet, and Maryann Feldman. 2011. "The mechanisms of collaboration in inventive teams: Composition, social networks, and geography." *Research Policy*, 40(1): 81–93.

Bernstein, Shai, Rebecca Diamond, Abhisit Jiranaphawiboon, Timothy McQuade, and Beatriz Pousada. 2022. "The contribution of high-skilled immigrants to innovation in the United States." NBER Working Paper 30797.

Borjas, George J. 1985. "Assimilation, changes in cohort quality, and the earnings of immigrants." *Journal of Labor Economics*, 3(4): 463–489.

Borjas, George J. 2003. "The labor demand curve is downward sloping: Reexamining the impact of immigration on the labor market." *The Quarterly Journal of Economics*, 118(4): 1335–1374.

Breschi, Stefano, Cornelia Lawson, Francesco Lissoni, Andrea Morrison, and Ammon Salter. 2020. "STEM migration, research, and innovation." *Research Policy*, 49(9): 104070.

Brown, J David, John S Earle, Mee Jung Kim, and Kyung Min Lee. 2019. "Start-ups, job creation, and founder characteristics." *Industrial and Corporate Change*, 28(6): 1637–1672.

Brown, J David, John S Earle, Mee Jung Kim, and Kyung Min Lee. 2020. "Immigrant entrepreneurs and innovation in the US high-tech sector." In *The Roles of Immigrants and Foreign Students in US Science, Innovation, and Entrepreneurship.* , ed. Ina Ganguli, Shulamit Kahn and Megan MacGarvie, 149–171. University of Chicago Press.

Card, David. 1990. "The impact of the Mariel boatlift on the Miami labor market." *ILR Review*, 43(2): 245–257.

Card, David. 2001. "Immigrant inflows, native outflows, and the local labor market impacts of higher immigration." *Journal of Labor Economics*, 19(1): 22–64.

Chiswick, Barry R, Yew Liang Lee, and Paul W Miller. 2005. "A longitudinal analysts of immigrant occupational mobility: A test of the immigrant assimilation hypothesis." *International Migration Review*, 39(2): 332–353.

Chiswick, Carmel U. 2009. "The economic determinants of ethnic assimilation." *Journal of Population Economics*, 22(4): 859–880.

Crown, Daniel, Alessandra Faggian, and Jonathan Corcoran. 2020. "Foreign-born graduates and innovation: Evidence from an Australian skilled visa program." *Research Policy*, 49(9): 103945.

Dreisigmeyer, David, Nathan Goldschlag, Marina Krylova, Wei Ouyang, Elisabeth Perlman, et al. 2018. "Building a better bridge: Improving patent assignee-firm links." Center for Economic Studies, U.S. Census Bureau CES Technical Notes Series 18-01.

Fairlie, Robert W, and Bruce D Meyer. 2003. "The effect of immigration on native self-employment." *Journal of Labor Economics*, 21(3): 619–650.

Fairlie, Robert W, and Magnus Lofstrom. 2015. "Immigration and entrepreneurship." In *Handbook of the Economics of International Migration*. Vol. 1, 877–911.

Ferrucci, Edoardo, and Francesco Lissoni. 2019. "Foreign inventors in Europe and the United States: Diversity and patent quality." *Research Policy*, 48(9): 103774.

Graham, Stuart JH, Cheryl Grim, Tariqul Islam, Alan C Marco, and Javier Miranda. 2018. "Business dynamics of innovating firms: Linking US patents with administrative data on workers and firms." *Journal of Economics & Management Strategy*, 27(3): 372–402.

Griffith, Rachel, Elena Huergo, Jacques Mairesse, and Bettina Peters. 2006. "Innovation and productivity across four European countries." *Oxford Review of Economic Policy*, 22(4): 483–498.

Hall, Bronwyn H. 2011. "Innovation and productivity." NBER Working Paper 17178.

Hart, David M, and Zoltan J Acs. 2011. "High-tech immigrant entrepreneurship in the United States." *Economic Development Quarterly*, 25(2): 116–129.

Hunt, Jennifer. 2011. "Which immigrants are most innovative and entrepreneurial? Distinctions by entry visa." *Journal of Labor Economics*, 29(3): 417–457.

Hunt, Jennifer, and Marjolaine Gauthier-Loiselle. 2010. "How much does immigration boost innovation?" *American Economic Journal: Macroeconomics*, 2(2): 31–56.

Kahn, Shulamit, and Megan J MacGarvie. 2016. "How important is US location for research in science?" *Review of Economics and Statistics*, 98(2): 397–414.

Kahn, Shulamit, Giulia La Mattina, and Megan J MacGarvie. 2017. ""Misfits," "stars," and immigrant entrepreneurship." *Small Business Economics*, 49(3): 533–

557.

Kerr, Sari Pekkala, and William R. Kerr. 2016. "Immigrant entrepreneurship." In *Measuring Entrepreneurial Businesses: Current Knowledge and Challenges*. 187–249. University of Chicago Press.

Kerr, Sari Pekkala, and William R. Kerr. 2020. "Immigrant entrepreneurship in America: Evidence from the survey of business owners 2007 & 2012." *Research Policy*, 49(3): 103918.

Kerr, William R. 2008. "The ethnic composition of US inventors." HBS Finance Working Paper 08-006.

Kerr, William R. 2017*a*. "Entrepreneurship in larger companies." In *The Entrepreneur's Roadmap: From Concept to IPO, edited by Tim Dempsey and Bonnie Hyun*. 161–164. Chicago, IL: Caxton Business Legal.

Kerr, William R. 2017*b*. "U.S. high-skilled immigration, innovation, and entrepreneurship: Empirical approaches and evidence." In *The International Mobility of Talent and Innovation: New Evidence and Policy Implications, Part II Intellectual Property, Innovation and Economic Development*. , ed. Carsten Fink and Ernest Miguelez, Chapter 6, 193–221. Cambridge University Press.

Kerr, William R., and William F Lincoln. 2010. "The supply side of innovation: H-1B visa reforms and US ethnic invention." *Journal of Labor Economics*, 28(3): 473–508.

Kim, Yoonha. 2019. "Preference vs Constraints: Generational Transmission of Immigrant Entrepreneurship." In *Academy of Management Proceedings*. Vol. 2019, 17479. Academy of Management Briarcliff Manor, NY 10510.

Laursen, Keld, Bart Leten, Ngoc Han Nguyen, and Mark Vancauteren. 2020. "Mounting corporate innovation performance: The effects of high-skilled migrant hires and integration capacity." *Research Policy*, 49(9): 104034.

Lee, Kyung Min, Mee Jung Kim, John S Earle, Lokesh Dani, Eric Childress, and J David Brown. 2022. *African-American entrepreneurs: Contributions and challenges.* Office of Advocacy, US Small Business Administration.

Levine, Ross, and Yona Rubinstein. 2017. "Smart and illicit: Who becomes an entrepreneur and do they earn more?" *The Quarterly Journal of Economics*, 132(2): 963–1018.

Mairesse, Jacques, and Pierre Mohnen. 2010. "Using innovation surveys for econometric analysis." In *Handbook of the Economics of Innovation*. Vol. 2, 1129–1155.

OECD/Eurostat. 2018. *Oslo manual 2018. Guidelines for collecting, reporting and using data on innovation, 4th edition.* Paris: OECD Publishing.

Ottaviano, Gianmarco IP, and Giovanni Peri. 2012. "Rethinking the effect of immi-

gration on wages." *Journal of the European Economic Association*, 10(1): 152–197.

Parisi, Maria Laura, Fabio Schiantarelli, and Alessandro Sembenelli. 2006. "Productivity, innovation and R&D: Micro evidence for Italy." *European Economic Review*, 50(8): 2037–2061.

Parker, Simon C. 2004. *The economics of self-employment and entrepreneurship.* Cambridge University Press.

Parrotta, Pierpaolo, Dario Pozzoli, and Mariola Pytlikova. 2014. "The nexus between labor diversity and firm's innovation." *Journal of Population Economics*, 27: 303–364.

Posch, Max, Jonathan Schulz, and Joseph Henrich. 2023. "Surname diversity, social ties and innovation." GMU Working Paper in Economics No. 23-31.

Saxenian, AnnaLee. 2002. "Silicon Valley's new immigrant high-growth entrepreneurs." *Economic Development Quarterly*, 16(1): 20–31.

Stephan, Paula E, and Sharon G Levin. 2001. "Exceptional contributions to US science by the foreign-born and foreign-educated." *Population Research and Policy Review*, 20(1): 59–79.

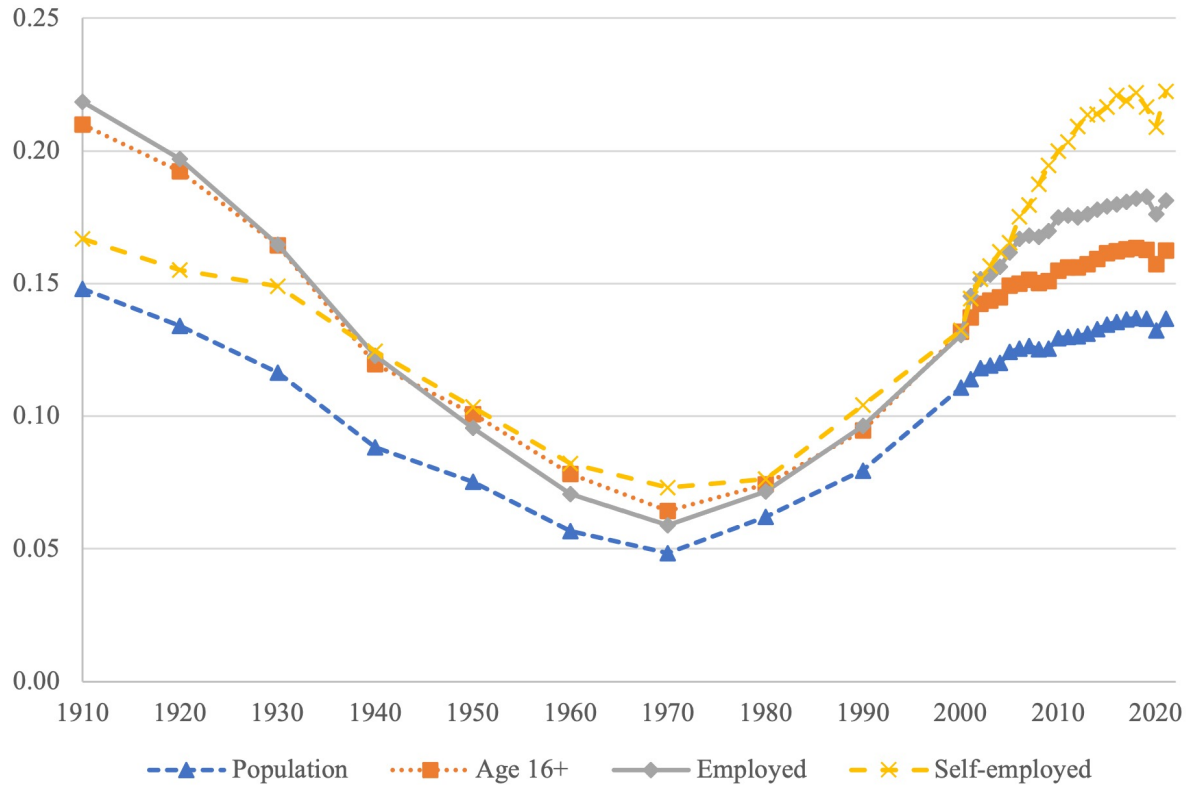
U.S. Patent and Trademarke Office Patent Technology Monitoring Team. 2014. "Patenting by Organizations (Utility Patents) 2014, PARTS A1, B." https://www.uspto.gov/web/offices/ac/ido/oeip/taf/topo_14.htm.

Wadhwa, Vivek, AnnaLee Saxenian, Ben A Rissing, and Gary Gereffi. 2007*a*. "America's new immigrant entrepreneurs: Part I." Duke Science, Technology & Innovation Paper No.23.

Wadhwa, Vivek, Ben A Rissing, AnnaLee Saxenian, and Gary Gereffi. 2007*b*. "Education, entrepreneurship and immigration: America's new immigrant entrepreneurs, Part II." Working Paper, Available at SSRN: https://ssrn.com/abstract=991327.

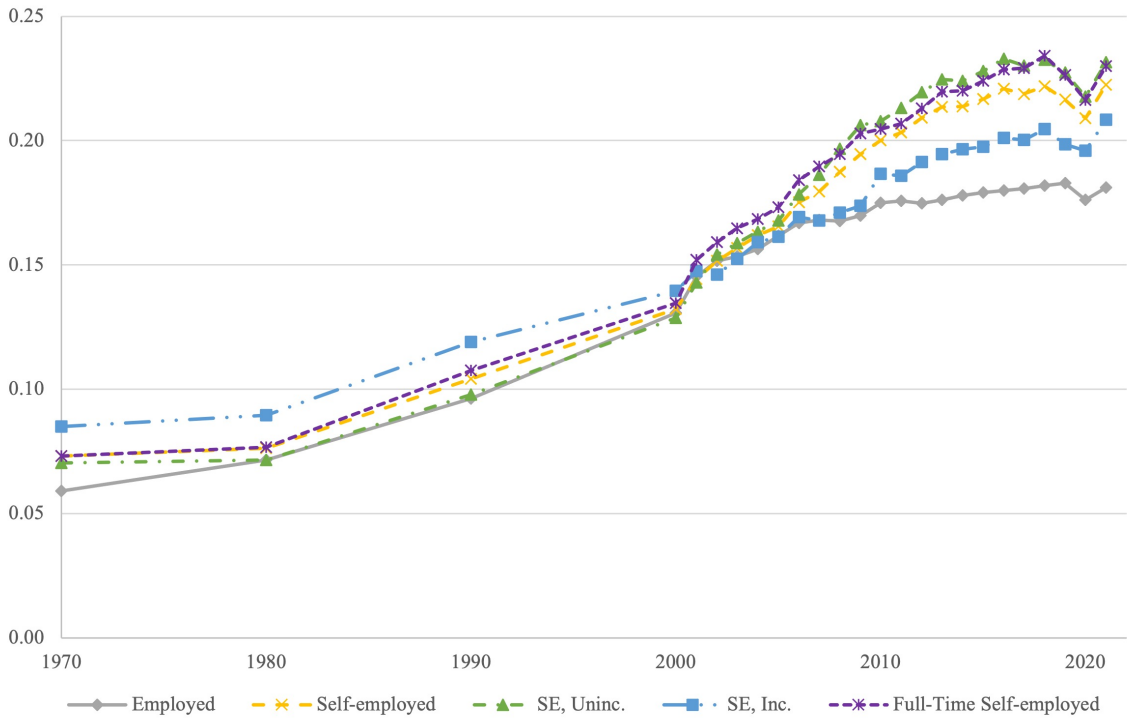
6 Figures and Tables

Figure 1: Share of Immigrants by Population Sub-Group, 1910-2021



Note: Data sources include the 1910-2000 Census and 2001-2021 American Community Survey. Estimates are weighted by final person weights provided by the Census Bureau. Consistent with the ASE, immigrants are defined as foreign-born people who are naturalized citizens or noncitizens, while those who were born in the U.S., born in U.S. outlying areas, or born abroad of American parents are defined as natives. The exceptions are 1910 and 1960, when the decennial census does not have complete citizenship information, and we use birthplace (foreign-born) in place of immigrant. The true immigrant shares for 1910 and 1960 could be slightly larger because of the small group of those born abroad with American parents who cannot be distinguished from immigrants who were not citizens at birth.

Figure 2: Share of Immigrants by Employment and Type of Self-Employment, 1970-2021



Note: Data sources include the 1970-2000 Census and the 2001-2021 American Community Survey. Estimates are weighted by final person weights provided by the Census Bureau. Naturalized citizens and noncitizens are defined as immigrants, while those who were born in the U.S., born in the U.S. outlying areas, or born abroad of American parents are natives.

Table 1a: Definitions and Sources of Innovation Measures

Variable	**Description**
Innovation count	Number of innovation activities (product or process)
Any innovation	Conducted any innovation activity (among those listed below)
Product innovation activities	
Any product innovation	Made any product innovation (among those listed below)
Completely new product	Sold a new good or service that no other business has ever offered before
New product for firm	Sold a new good or service that this business has never offered before
Improved performance	Improved a good or service's performance by making changes in materials, equipment, software, or other
New use	Developed a new use for a good or service
New feature	Added a new feature to a good or service
Easy to use	Made it easier for customers to use a good or service
Process innovation activities	
Any process innovation	Made any process innovation (among those listed below)
New support activity	Applied a new way of purchasing, accounting, computing, maintenance, inventory control, or other support activity
Distribution improvement	Reduced costs by changing the way a good or service was distributed
Upgraded technique	Upgraded a technique, equipment, or software to significantly improve a good or service
Process improvement	Made a significant improvement in a technique or process by increasing automation, decreasing energy consumption, or using better software
Decreased production cost	Decreased production costs by improving the materials, software, or other components
Delivery improvement	Changed a delivery method to be faster or more reliable
R&D Activities	
Basic R&D	Produced findings that could be published in academic journals or presented at scientific conferences, created new scientific research or technical solutions that can be generalized to other situations, conducted work to discover previously unknown scientific facts, structures, or relationships, or conducted work to extend the understanding of scientific facts, relationships or principles in a way that could be useful to others
Applied R&D	Conducted work that might lead to a patent, developed and tested prototypes that were derived from scientific research or technical findings, or applied scientific or technical knowledge in a way that has never been done before

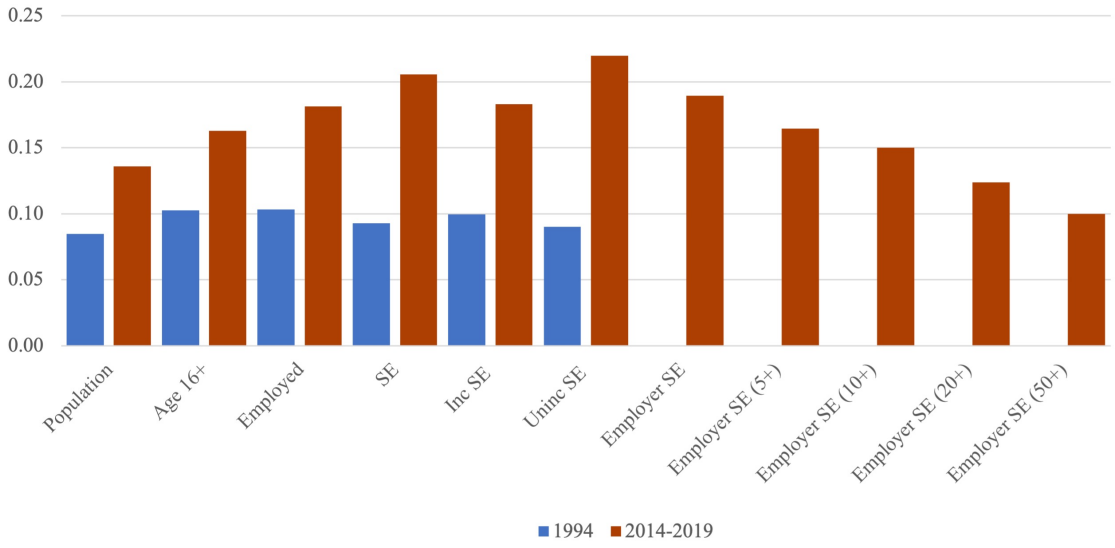
Note: All innovation and R&D activities are measured for the period 2012-2014 from the 2014 Annual Survey of Entrepreneurs (ASE). Innovation count, any innovation, any product innovation, and any process innovation are constructed based on the reported activities, and the R&D variables are each constructed based on six different types of R&D, respectively.

Table 1b: Definitions and Sources of Intellectual Property, Employment, and Productivity Measures

Variable	**Description**	**Source**
Intellectual Property		
Copyright or trademark	Owned copyright or trademark in 2014	ASE
Patent pending	Applied for a patent(pending) in 2014	ASE
Patent owned	Owned a patent(granted) in 2014	ASE
Patent>0 2012-14	Received at least 1 patent from 2012 to 2014	CBPCD
Patent>2 2012-14	Received at least 3 patents from 2012 to 2014	CBPCD
Patent>0 2015-17	Received at least 1 patent from 2015 to 2017	CBPCD
Patent>2 2015-17	Received at least 3 patents from 2015 to 2017	CBPCD
Employment & Productivity in 2014		
Log employment	Log (employees)	BR
Log productivity	Log (revenue/employees) - 4-digit industry mean	BR
Employment top 5%	Above 95th percentile of employment	BR
Productivity top 5%	Above 95th percentile of productivity	BR

Note: Intellectual property variables from the ASE are measured for the period 2012-2014. ASE = 2014 Annual Survey of Entrepreneurs; BR = Business Register; CBPCD = Census Bureau Patent Crosswalk Data.

Figure 3: Share of Immigrants in Population, Self-Employment, and Employer Size Groups



Note: Data sources are Current Population Survey (CPS) Monthly for 1994 and CPS Monthly Outgoing Rotation Groups (ORGs) for 2014-2019. The table reports the share of foreign born for each sample. The numbers are weighted by CPS final person weights. SE = Self-employed; Inc SE = Incorporated self-employed; Uninc SE = Unincorporated self-employed. Naturalized citizens and noncitizens are defined as immigrants, while those who were born in the U.S., born in the U.S. outlying areas, or born abroad of American parents are natives.

Table 2: Descriptive Statistics: Firm and Owner Characteristics by Ownership Type

Variables	Any Immigrant	Only Immigrant	Diverse Native-Immigrant	Diverse Within-Immigrant	Only Native
Firm Age (years)					
0-2	22.2	22.7	19.5	26.9	14.7
3-5	21.1	21.7	18.0	18.5	14.5
6-10	24.2	24.7	22.0	22.7	20.7
11-15	31.0	29.5	37.8	30.1	47.0
16+	1.6	1.3	2.7	1.7	3.2
Number of Owners					
1	55.3	67.4	0.0	0.0	59.2
2-4	39.8	29.0	88.6	84.0	36.9
2	32.3	25.2	64.8	57.7	30.6
3	4.8	2.6	15.0	16.1	4.4
4	2.6	1.2	8.8	10.2	1.8
5+	4.2	2.8	10.7	14.9	3.5
Gender					
Any female	44.2	39.0	67.9	62.8	40.7
Only male	55.8	61.0	32.1	37.2	59.3
Only female	17.8	20.9	3.9	3.8	16.6
Mixed gender - unrelated	5.8	3.7	15.4	24.1	3.7
Mixed gender - family	20.6	14.5	48.6	35.0	20.4
Owner Age					
<25	0.4	0.3	0.6	0.2	0.3
25-34	6.3	6.2	6.8	5.8	5.2
35-44	23.9	24.5	21.0	25.3	15.9
45-54	32.4	33.1	29.4	34.8	28.5
55-64	25.0	24.8	26.2	22.4	31.5
65+	12.0	11.1	15.9	11.4	18.6
Education					
Less than high school	8.7	9.6	4.8	4.7	2.4
High school	18.9	19.7	15.1	15.3	18.8
Some college	22.1	21.7	23.9	22.5	27.5
Bachelor's degree	24.5	23.2	30.6	24.1	28.1
Graduate	25.8	25.8	25.7	33.3	23.2
Experience					
Prior business	36.0	35.4	38.9	44.4	31.6
Veteran	2.4	1.5	6.5	2.7	11.3
Observations in ASE Sample	31,000	24,600	6,400	750	168,000
Weighted observations	1,021,000	837,000	184,000	23,000	4,320,000

Note: Data are firm-level, from the 2014 Annual Survey of Entrepreneurs (ASE). The table reports the percentages for each variable by ownership type: (1) any immigrant owner (Any Immigrant), (2) only immigrant owners (Only Immigrant), (3) mixed immigrant and native owners (Diverse Native-Immigrant), (4) with differences in ethnicity or race among immigrant owners (Diverse Within-Immigrant), and (5) only native owners (Only Native). The numbers are weighted by the ASE regression sampling weights and may not sum to 100% because of rounding or "don't know" responses. These results are approved for dissemination by the DRB (CBDRB-FY2020-CES005-034; CBDRB-FY23-CES019-010).

Table 3: Descriptive Statistics: Firm Characteristics by Ownership Type

Variables	Any Immigrant	Only Immigrant	Diverse Native-Immigrant	Diverse Within-Immigrant	Only Native
Start-up Finance Amount					
No capital needed	5.1	5.1	5.1	4.3	9.2
<5k	11.4	11.6	10.5	9.5	16.1
5k-10k	8.1	8.4	6.5	5.8	8.6
10k-25k	12.7	13.2	10.5	8.7	12.1
25k-50k	10.9	11.3	9.2	11.4	9.3
50k-100k	12.8	12.9	12.6	12.9	9.9
100k-250k	13.4	13.3	13.7	15.2	9.6
250k-1m	8.5	7.9	11.2	11.0	6.4
1m-3m	1.8	1.5	3.3	3.1	1.4
3m more	0.7	0.6	1.3	1.6	0.5
Start-up Finance Source(s)					
Savings	75.4	75.6	74.7	75.8	67.3
Assets	10.4	10.0	12.1	12.2	10.0
Home equity	8.2	7.9	9.1	12.0	7.2
Personal credit card(s)	11.5	11.5	11.6	13.6	10.5
Business credit card(s)	5.1	4.9	6.0	6.9	5.6
SBA loan	1.7	1.5	2.7	1.7	2.0
Bank loan	14.2	13.7	16.3	15.4	18.7
Government loan	0.4	0.4	0.5	D	0.4
Family loan	4.8	4.7	4.9	4.6	5.1
Outside investor	0.6	0.4	1.4	2.0	0.5
Grants	0.3	0.2	0.6	D	0.2
Other source	3.7	3.4	4.7	3.2	3.3
Industry					
Primary sector	0.3	0.2	0.6	0.0	1.1
Construction	6.8	6.6	7.5	5.0	13.5
Manufacturing	3.5	3.0	6.2	5.6	4.7
Wholesale trade	6.8	6.7	6.9	7.5	5.2
Retail trade	15.6	16.5	11.5	11.8	10.8
Transportation	3.2	3.3	2.6	3.6	3.0
Information and finance	2.9	2.6	4.3	4.6	6.1
Real estate	3.6	3.2	4.9	5.1	5.1
Professional	12.3	11.7	15.0	15.3	16.7
Administrative and support	4.6	4.5	5.1	3.1	6.5
Education	1.1	1.0	1.6	D	1.1
Health	13.5	13.9	11.2	16.6	10.9
Art and entertainment	0.8	0.7	1.3	D	1.9
Accommodation and food	16.9	17.3	14.9	14.4	6.2
Other services	8.2	8.7	6.2	6.1	6.6
High-tech sector	6.4	6.0	7.8	13.0	4.9
Observations in ASE Sample	31,000	24,600	6,400	750	168,000
Weighted observations	1,021,000	837,000	184,000	23,000	4,320,000

Note: Data are firm-level, from the 2014 Annual Survey of Entrepreneurs (ASE). The table reports the percentages of firms for each variable by ownership type: (1) any immigrant owner (Any Immigrant), (2) only immigrant owners (Only Immigrant), (3) mixed immigrant and native owners (Diverse Native-Immigrant), (4) with differences in ethnicity or race among immigrant owners (Diverse Within-Immigrant), and (5) only native owners (Only Native). The numbers are weighted by the ASE regression sampling weights and may not sum to 100% because of rounding or "don't know" responses. These results are approved for dissemination by the DRB (CBDRB-FY2020-CES005-034; CBDRB-FY23-CES019-010). The cells with "D" indicate the numbers cannot be disclosed due to the limited number of observations.

Table 4a: Innovation Measures by Ownership Type

Variables	(1) Any Immigrant	(2) Only Immigrant	(3) Diverse Native-Immigrant	(4) Diverse Within-Immigrant	(5) Only Native
Innovation count	2.5	2.5	2.8	3.2	2.1
Any innovation	56.4	55.5	60.5	64.7	52.7
Product Innovation					
Any product innovation	44.5	44.0	47.2	51.5	39.7
Completely new product	7.2	6.8	8.7	10.3	4.7
New product for firm	14.5	13.4	19.3	18.8	15.4
Improved performance	29.7	28.9	33.3	34.5	26.2
New use	14.7	14.8	14.3	18.3	7.9
New feature	23.1	22.4	26.0	30.6	18.6
Easy to use	31.6	31.6	31.7	37.5	25.4
Process Innovation					
Any process innovation	46.3	45.4	50.7	54.7	43.2
New support activity	22.4	21.6	26.0	28.5	19.5
Distribution improvement	21.4	21.1	22.6	25.3	14.7
Upgraded technique	34.6	33.4	39.7	45.1	33.7
Process improvement	21.2	20.2	25.5	28.1	19.7
Decreased production cost	16.1	15.7	17.9	22.8	12.9
Delivery improvement	15.7	15.4	16.7	22.3	12.1
R&D Activities					
Basic R&D	5.6	5.0	8.4	12.1	3.8
Applied R&D	5.4	4.9	7.4	10.0	3.7
Observations in ASE Sample	31,000	24,600	6,400	750	168,000
Weighted observations	1,021,000	837,000	184,000	23,000	4,320,000

Note: Data are firm-level, from the 2014 Annual Survey of Entrepreneurs (ASE). The table reports the percentages of firms for dummy variables and means for continuous variables by ownership type: (1) any immigrant owner (Any Immigrant), (2) only immigrant owners (Only Immigrant), (3) mixed immigrant and native owners (Diverse Native-Immigrant), (4) with differences in ethnicity or race among immigrant owners (Diverse Within-Immigrant), and (5) only native owners (Only Native). All variables are dummy variables except for innovation count. The numbers are weighted by the ASE regression sampling weights and may not sum to 100% because of rounding or "don't know" responses. These results are approved for dissemination by the DRB (CBDRB-FY2020-CES005-034; CBDRB-FY23-CES019-010).

Table 4b: Intellectual Property, Employment, and Productivity by Ownership Type

Variables	(1) Any Immigrant	(2) Only Immigrant	(3) Diverse Native-Immigrant	(4) Diverse Within-Immigrant	(5) Only Native
Intellectual Property					
Copyright or trademark	8.3	7.0	14.3	13.3	8.0
Patent pending	1.1	0.8	2.5	4.9	0.7
Patent owned	1.4	1.0	2.9	4.4	1.1
Patent >0 2012-14	0.5	0.4	0.9	2.2	0.3
Patent >2 2012-14	0.2	0.1	0.3	1.1	0.1
Patent >0 2015-17	0.5	0.4	1.2	2.8	0.3
Patent >2 2015-17	0.2	0.1	0.3	D	0.1
Observations in ASE Sample	31,000	24,600	6,400	750	168,000
Weighted observations	1,021,000	837,000	184,000	23,000	4,320,000
Employment & Productivity					
Log employment	1.3	1.2	1.7	1.7	1.4
Log productivity	0.1	0.1	0.1	0.1	0.0
Employment top 5%	3.5	2.6	7.9	7.7	5.5
Productivity top 5%	6.9	7.0	6.6	6.8	4.6
Observations in ASE+LBD Sample	20,000	15,400	4,600	500	116,000
Weighted observations	1,006,000	813,000	193,000	16,500	4,331,000

Note: Data are firm-level, from the 2014 Annual Survey of Entrepreneurs (ASE), in the bottom panel linked to the Business Register (BR) and Census Bureau Patent Crosswalk Data (CBPCD). The table reports the percentages of firms for dummy variables and means for continuous variables by ownership type: (1) any immigrant owner (Any Immigrant), (2) only immigrant owners (Only Immigrant), (3) mixed immigrant and native owners (Diverse Native-Immigrant), (4) with differences in ethnicity or race among immigrant owners (Diverse Within-Immigrant), and (5) only native owners (Only Native). All variables are dummy variables except for employment and productivity. The numbers are weighted by ASE sampling weights and may not sum to 100% because of rounding or "don't know" responses. These results are approved for dissemination by the DRB (CBDRB-FY2020-CES005-034; CBDRB-FY23-CES019-010). The cell with "D" indicates the number cannot be disclosed due to the limited number of observations.

Table 5a: Impact of Any Immigrant Owner on Innovation Activities

Dependent Variables	(1) Native Mean	(2) Base	(3) + Finance	(4) + Industry
Innovation count	2.1	0.372	0.290	0.224
		(0.022)	(0.022)	(0.023)
Any innovation	52.7	3.414	1.806	1.304
		(0.356)	(0.354)	(0.363)
Product Innovation				
Any product innovation	39.7	4.297	2.929	1.629
		(0.356)	(0.356)	(0.364)
Completely new product	4.7	2.127	1.989	1.350
		(0.180)	(0.181)	(0.186)
New product for firm	15.4	-1.591	-2.283	-3.787
		(0.254)	(0.256)	(0.263)
Improved performance	26.2	3.274	2.298	1.693
		(0.326)	(0.327)	(0.335)
New use	7.9	6.296	5.987	5.258
		(0.246)	(0.246)	(0.252)
New feature	18.6	3.905	3.142	1.847
		(0.299)	(0.299)	(0.307)
Easy to use	25.4	5.713	4.747	3.700
		(0.331)	(0.331)	(0.341)
Process Innovation				
Any process innovation	43.2	2.934	1.441	1.469
		(0.357)	(0.356)	(0.366)
New support activity	19.5	2.186	1.321	0.764
		(0.298)	(0.298)	(0.310)
Distribution improvement	14.7	6.214	5.499	4.766
		(0.290)	(0.290)	(0.299)
Upgraded technique	33.7	0.908	-0.317	0.022
		(0.339)	(0.339)	(0.349)
Process improvement	19.7	1.684	1.001	1.069
		(0.291)	(0.292)	(0.302)
Decreased production cost	12.9	3.127	2.608	2.669
		(0.261)	(0.262)	(0.269)
Delivery improvement	12.1	3.381	2.960	3.065
		(0.259)	(0.260)	(0.268)
R&D Activities				
Basic R&D	3.8	1.656	1.609	1.236
		(0.161)	(0.162)	(0.164)
Applied R&D	3.7	1.435	1.411	1.098
		(0.158)	(0.159)	(0.163)

Note: The first column reports unconditional means (percentages for dummy variables and the mean for the continuous variable Innovation count) of firms with only native owners (Native Mean). In the other columns, each cell of the table reports the coefficient from a separate linear regression of the innovation activity on Any Immigrant owner. Robust standard errors are in parentheses. Coefficients and standard errors are multiplied by 100 for ease of reading. The "base" specification includes firm age, demographic characteristics (gender, age, and race/ethnicity), ownership team characteristics (size and family relationships), and human capital (education, veteran, and prior business experience). The second ("+ Finance") specification adds start-up capital amount categories, and the third ("+ Industry") additionally includes 4-digit industry. Results are weighted by ASE sampling weights. Observations = 199,000 firms. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004).

Table 5b: Impact of Any Immigrant Owner on Intellectual Property, Employment, and Productivity

Dependent Variables	(1) Native Mean	(2) Base	(3) + Finance	(4) + Industry
Intellectual Property				
Copyright or trademark	8.0	0.346	0.148	-0.584
		(0.197)	(0.198)	(0.202)
Patent pending	0.7	0.302	0.278	0.190
		(0.071)	(0.071)	(0.073)
Patent owned	1.1	0.380	0.357	0.233
		(0.081)	(0.082)	(0.083)
Patent>0 2012-14	0.3	0.182	0.179	0.141
		(0.046)	(0.047)	(0.048)
Patent>2 2012-14	0.1	0.080	0.080	0.060
		(0.026)	(0.026)	(0.026)
Patent>0 2015-17	0.3	0.211	0.209	0.180
		(0.050)	(0.050)	(0.051)
Patent>2 2015-17	0.1	0.089	0.090	0.080
		(0.029)	(0.029)	(0.030)
Employment & Productivity				
Log employment	1.4	-0.067	-0.110	-0.141
		(0.009)	(0.009)	(0.009)
Log productivity	0.0	0.103	0.094	0.107
		(0.009)	(0.009)	(0.009)
Employment top 5%	5.5	-1.102	-1.213	-1.317
		(0.155)	(0.155)	(0.162)
Productivity top 5%	4.6	1.858	1.800	1.908
		(0.223)	(0.224)	(0.231)

Note: The first column reports unconditional means (percentages for dummy variables and the mean for the continuous variables) of only native owners (Native Mean). In the other columns, each cell of the table reports the coefficient from a separate linear regression of the innovation activity on Any Immigrant owner. Robust standard errors are in parentheses. Coefficients and standard errors are multiplied by 100 for ease of reading. The "base" specification includes firm age, demographic characteristics (gender, age, and race/ethnicity), ownership team characteristics (size and family relationships), and human capital (education, veteran, and prior business experience). The second ("+ Finance") specification adds start-up capital amount categories, and the third ("+ Industry") additionally includes 4-digit industry. Results are weighted by ASE sampling weights. Observations = 199,000 firms for the Intellectual Property regressions and 135,000 firms for the Employment and Productivity regressions. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004).

Table 6: Impact of Immigrant Status on Probabilities of Self-Employment, Employer, and Employer Size Groups

Dependent Variables	(1) Native Mean	(2) No Controls	(3) + Controls	(4) + Industry
Sample: Employed				
Self-employed	10.8	1.649	0.576	-0.637
		(0.052)	(0.054)	(0.052)
Self-employed Incorporated	4.3	0.000	-0.359	-0.751
		(0.032)	(0.034)	(0.034)
Sample: Self-employed				
Employer Self-employed	24.3	-2.355	-2.059	-3.812
		(0.377)	(0.393)	(0.383)
5+ employees	10.6	-2.546	-2.255	-3.507
		(0.251)	(0.266)	(0.263)
10+ employees	6.0	-1.926	-1.728	-2.561
		(0.185)	(0.197)	(0.198)
20+ employees	2.9	-1.320	-1.222	-1.627
		(0.119)	(0.127)	(0.132)
50+ employees	0.9	-0.534	-0.454	-0.587
		(0.062)	(0.065)	(0.069)

Note: Data from the Current Population Survey (CPS) Outgoing Rotation Groups (ORGs) from 2014 to 2019. For the self-employed regressions, the sample is all employed; for the employer regressions, the sample is restricted to all self-employed. The first column reports the unconditional mean for native owners, and each cell of the other columns (2-4) reports the coefficient on immigrant status from a separate regression, all expressed in percentage terms. Robust standard errors are in parentheses. Coefficients and standard errors are multiplied by 100 for ease of reading. The "No Controls" specification includes no control variables. The second ("+ Controls") specification includes individual demographic (female, age, and married), and human capital variables (education and veteran). The third ("+ Industry") specification includes 2-digit industry dummies. Results are weighted by CPS individual weights. For self-employed regressions, observations = 3,611,965 for the "No Controls" specifications and 3,594,601 for the rest. For employer regressions, observations = 105,086 for the "No Controls" specifications and 104,976 for the rest.

Table 7a: Impact of Any Immigrant Owner on Innovation Activities by High & Low-Tech Sector

	High-Tech		Low-Tech	
Dependent Variables	Native Mean	Base	Native Mean	Base
Completely new product	11.2	0.701	4.3	2.055
		(1.023)		(0.180)
New product for firm	22.9	-1.279	15.0	-1.735
		(1.296)		(0.258)
Other product innovation	52.0	3.558	35.9	4.857
		(1.515)		(0.363)
Process innovation	57.7	2.152	42.5	2.875
		(1.501)		(0.367)
Basic R&D	17.7	2.973	3.1	1.198
		(1.246)		(0.149)
Applied R&D	14.9	2.773	3.1	1.029
		(1.190)		(0.148)

Note: The first and the third columns report the unconditional mean (in percentage) of only native owners (Only Native). In the other columns, each cell reports the coefficient from a separate LPM regression of the Innovation activity dependent variables on Any Immigrant owner in the high-tech and low-tech sectors. Robust standard errors are in parentheses. The base specification includes firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). Coefficients and standard errors are multiplied by 100 for ease of reading. The numbers are weighted by ASE sampling weights. Observations for high-tech and low-tech samples are 8,900 and 190,000 firms, respectively. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004).

Table 7b: Impact of Any Immigrant Owner by High & Low-Tech Sector: Intellectual Property, Employment and Productivity

	High-Tech		Low-Tech	
Dependent Variables	Native Mean	Base	Native Mean	Base
Intellectual Property				
Copyright or trademark	19.7	-2.560	7.4	0.319
		(1.159)		(0.196)
Patent pending	3.5	1.495	0.6	0.116
		(0.676)		(0.061)
Patent owned	4.2	1.017	0.9	0.246
		(0.659)		(0.075)
Patent>0 2012-14	1.6	1.160	0.2	0.085
		(0.450)		(0.039)
Patent>2 2012-14	0.5	0.832	0.1	0.019
		(0.305)		(0.019)
Patent>0 2015-17	1.5	1.556	0.2	0.079
		(0.492)		(0.041)
Patent>2 2015-17	0.6	0.515	0.1	0.045
		(0.297)		(0.023)
Employment & Productivity				
Log employment	1.2	-0.004	1.4	-0.066
		(0.036)		(0.009)
Log productivity	0.0	0.079	0.0	0.105
		(0.037)		(0.009)
Employment top 5%	5.1	0.716	5.5	-1.232
		(0.689)		(0.159)
Productivity top 5%	3.6	3.089	4.6	1.773
		(0.889)		(0.231)

Note: The first and the third columns report the unconditional mean (percentages for dummy variables and means for continuous variables) of only native owners (Only Native). In the other columns, each cell reports the coefficient from a separate LPM regression of the Intellectual Property, Employment and Productivity dependent variables on Any Immigrant owner in the high-tech and low-tech sectors. Robust standard errors are in parentheses. The base specification includes firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). Coefficients and standard errors are multiplied by 100 for ease of reading. Results are weighted by ASE sampling weights. Observations for high-tech and low-tech samples for the Intellectual Property regressions are 8,900 and 190,000 firms, respectively, and 6,700 and 129,000 firms for the Employment and Productivity regressions. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004).

Table 8a: Immigrant-Native Diversity and Innovation Activities

Dependent Variables	Native Mean	Base		
		Only Immigrant	Diverse Native-Immigrant	Diverse Within-Immigrant
Innovation count	2.1	0.363	0.363	0.351
		(0.025)	(0.051)	(0.151)
Any innovation	52.7	3.412	2.912	3.368
		(0.393)	(0.753)	(2.079)
Product Innovation				
Any product innovation	39.7	4.451	3.170	2.912
		(0.393)	(0.766)	(2.171)
Completely new product	4.7	1.966	2.593	1.766
		(0.197)	(0.432)	(1.364)
New product for firm	15.4	-2.133	0.599	1.397
		(0.274)	(0.604)	(1.723)
Improved performance	26.2	3.027	4.112	1.720
		(0.359)	(0.721)	(2.075)
New use	7.9	6.600	4.675	2.068
		(0.275)	(0.534)	(1.711)
New feature	18.6	3.767	3.972	4.138
		(0.328)	(0.670)	(2.019)
Easy to use	25.4	6.245	2.961	2.654
		(0.366)	(0.714)	(2.118)
Process Innovation				
Any process innovation	43.2	2.848	2.811	3.379
		(0.393)	(0.768)	(2.157)
New support activity	19.5	2.018	2.547	2.368
		(0.325)	(0.673)	(1.968)
Distribution improvement	14.7	6.470	4.930	0.955
		(0.320)	(0.641)	(1.905)
Upgraded technique	33.7	0.378	2.412	6.503
		(0.372)	(0.748)	(2.165)
Process improvement	19.7	1.391	2.505	2.859
		(0.317)	(0.664)	(1.951)
Decrease production cost	12.9	3.194	2.267	3.860
		(0.286)	(0.585)	(1.861)
Delivery improvement	12.1	3.386	2.745	4.839
		(0.285)	(0.573)	(1.832)
R&D Activities				
Basic R&D	3.8	1.160	3.241	4.827
		(0.171)	(0.412)	(1.398)
Applied R&D	3.7	1.004	2.917	3.632
		(0.170)	(0.387)	(1.275)

Note: The first column reports the unconditional mean (percentages for dummy variables and means for continuous variables) of only native owners (Only Native). In the other columns, each cell of the table reports the coefficient from a separate linear regression of the innovation activities dependent variable on Only, Diverse Native-, and Within-Immigrant owner. Robust standard errors are in parentheses. The base specification includes firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). Coefficients and standard errors are multiplied by 100 for ease of reading. The numbers are weighted by ASE sampling weights. Observations = 199,000 firms. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004; CBDRB-FY23-CES019-010).

Table 8b: Immigrant-Native Diversity, Intellectual Property, Employment, and Productivity

Dependent Variables	Native Mean	Base		
		Only Immigrant	Diverse Native-Immigrant	Diverse Within-Immigrant
Intellectual Property				
Copyright or trademark	8.0	-0.433	3.486	1.683
		(0.205)	(0.527)	(1.432)
Patent pending	0.7	0.031	1.117	3.047
		(0.070)	(0.228)	(0.936)
Patent owned	1.1	0.146	1.116	2.153
		(0.080)	(0.245)	(0.891)
Patent>0 2012-14	0.3	0.125	0.253	1.371
		(0.048)	(0.131)	(0.643)
Patent>2 2012-14	0.1	0.042	0.139	0.837
		(0.028)	(0.076)	(0.423)
Patent>0 2015-17	0.3	0.084	0.540	1.825
		(0.051)	(0.153)	(0.712)
Patent>2 2015-17	0.1	0.061	0.097	D
		(0.033)	(0.074)	D
Employment & Productivity				
Log Employment	1.4	-0.091	0.009	0.017
		(0.010)	(0.020)	(0.051)
Log Productivity	0.0	0.109	0.081	-0.040
		(0.010)	(0.018)	(0.052)
Employment top 5%	5.5	-1.315	-0.484	-0.092
		(0.152)	(0.436)	(1.151)
Productivity top 5%	4.6	2.038	1.220	-1.014
		(0.250)	(0.474)	(1.304)

Note: The first column reports the unconditional mean (percentages for dummy variables and means for continuous variables) of only native owners (Only Native). In the other columns, each cell of the table reports the coefficient from a separate linear regression of the innovation activities dependent variable on Only, Diverse Native-, and Diverse Within-Immigrant owner. Robust standard errors are in parentheses. The base specification includes firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). Coefficients and standard errors are multiplied by 100 for ease of reading. The numbers are weighted by ASE sampling weights. Observations = 199,000 firms for Intellectual Property and 135,000 firms for Employment and Productivity regressions. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004; CBDRB-FY23-CES019-010). The cell with "D" indicates the number cannot be disclosed due to the limited number of observations.

Appendix A

Table A.1: Impact of Any Immigrant Owner on Innovation Activities by High & Low-Tech Sector

	(1) High-Tech		(2) Low-Tech	
Dependent Variables	+ Finance	+ Industry	+ Finance	+ Industry
Completely new product	0.450	-0.722	1.903	1.472
	(1.018)	(1.026)	(0.181)	(0.187)
New product for firm	-1.544	-3.890	-2.505	-3.717
	(1.291)	(1.299)	(0.259)	(0.267)
Other product innovation	2.838	-0.800	3.405	3.024
	(1.502)	(1.495)	(0.363)	(0.374)
Process innovation	1.759	-0.262	1.184	1.670
	(1.486)	(1.490)	(0.367)	(0.378)
Basic R&D	2.604	1.019	1.116	1.182
	(1.233)	(1.204)	(0.150)	(0.155)
Applied R&D	2.476	1.978	0.978	0.950
	(1.184)	(1.155)	(0.149)	(0.156)

Note: Each cell of the table reports the coefficient from a separate LPM regression of the innovation activity dependent variable on Any Immigrant owner in high-tech and low-tech sector. Robust standard errors are in parentheses. All specifications include firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). The first ("+ Finance") specification includes start-up capital amount categories, and the second ("+ Industry") additionally includes 4-digit industry. Coefficients and standard errors are multiplied by 100 for ease of reading. The numbers are weighted by ASE sampling weights. Observations for high-tech and low-tech samples are 8,900 and 190,000 firms, respectively. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004).

Table A.2: Impact of Any Immigrant Owner by High & Low-Tech Sector: Intellectual Property, Employment and Productivity

	(1) High-Tech		(2) Low-Tech	
Dependent Variables	+ Finance	+ Industry	+ Finance	+ Industry
Intellectual Property				
Copyright or trademark	-2.728	-4.309	0.082	-0.328
	(1.157)	(1.160)	(0.197)	(0.202)
Patent pending	1.334	1.070	0.090	0.095
	(0.663)	(0.650)	(0.061)	(0.064)
Patent owned	1.013	0.491	0.221	0.200
	(0.656)	(0.644)	(0.076)	(0.078)
Patent>0 2012-14	1.175	1.005	0.079	0.083
	(0.450)	(0.449)	(0.040)	(0.041)
Patent>2 2012-14	0.860	0.776	0.018	0.014
	(0.309)	(0.308)	(0.019)	(0.021)
Patent>0 2015-17	1.553	1.478	0.079	0.084
	(0.483)	(0.483)	(0.041)	(0.043)
Patent>2 2015-17	0.527	0.507	0.047	0.049
	(0.289)	(0.295)	(0.024)	(0.025)
Employment & Productivity				
Log employment	0.021	0.025	-0.115	-0.152
	(0.035)	(0.035)	(0.009)	(0.009)
Log productivity	0.082	0.085	0.094	0.108
	(0.037)	(0.037)	(0.009)	(0.009)
Employment top 5%	1.038	0.611	-1.372	-1.443
	(0.684)	(0.688)	(0.160)	(0.167)
Productivity top 5%	3.061	3.052	1.732	1.817
	(0.881)	(0.870)	(0.232)	(0.240)

Note: Each cell of the table reports the coefficient from a separate linear regression of the intellectual property, employment, and productivity dependent variable on Any Immigrant owner in high-tech and low-tech sector. Robust standard errors are in parentheses. All specifications include firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). The first ("+ Finance") specification includes start-up capital amount categories, and the second ("+ Industry") additionally includes 4-digit industry. Coefficients and standard errors are multiplied by 100 for ease of reading. Results are weighted by ASE sampling weights. Observations for high-tech and low-tech samples for Intellectual Property are 8,900 and 190,000 firms, respectively, and 6,700 and 129,000 for Employment and Productivity. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004).

Table A.3: Immigrant-Native Diversity and Innovation Activities

	(1) + Finance			(2) + Industry		
Dependent Variables	Only Immigrant	Diverse Native-Immigrant	Diverse Within-Immigrant	Only Immigrant	Diverse Native-Immigrant	Diverse Within-Immigrant
Innovation count	0.274	0.308	0.370	0.214	0.227	0.312
	(0.025)	(0.051)	(0.150)	(0.025)	(0.049)	(0.147)
Any innovation	1.668	1.894	3.740	1.279	0.994	2.972
	(0.392)	(0.750)	(2.076)	(0.403)	(0.735)	(2.052)
Product Innovation						
Any product innovation	2.977	2.268	3.208	1.711	0.915	2.703
	(0.392)	(0.764)	(2.165)	(0.404)	(0.742)	(2.112)
Completely new product	1.822	2.485	1.778	1.205	1.783	1.295
	(0.197)	(0.431)	(1.362)	(0.204)	(0.421)	(1.337)
New product for firm	-2.881	0.145	1.563	-4.526	-0.914	1.642
	(0.275)	(0.605)	(1.708)	(0.285)	(0.578)	(1.686)
Improved performance	1.974	3.467	1.961	1.484	2.395	1.177
	(0.359)	(0.722)	(2.070)	(0.370)	(0.701)	(2.058)
New use	6.271	4.458	2.108	5.596	3.658	1.538
	(0.275)	(0.533)	(1.707)	(0.283)	(0.522)	(1.677)
New feature	2.951	3.442	4.279	1.685	2.041	3.930
	(0.328)	(0.668)	(2.010)	(0.339)	(0.651)	(1.981)
Easy to use	5.207	2.321	2.853	4.191	1.340	2.359
	(0.366)	(0.712)	(2.107)	(0.378)	(0.700)	(2.056)
Process Innovation						
Any process innovation	1.229	1.863	3.739	1.412	1.321	2.774
	(0.393)	(0.765)	(2.145)	(0.405)	(0.751)	(2.122)
New support activity	1.082	1.999	2.601	0.473	1.633	2.575
	(0.325)	(0.672)	(1.960)	(0.340)	(0.669)	(1.966)
Distribution improvement	5.701	4.457	1.110	4.946	3.883	1.187
	(0.320)	(0.640)	(1.899)	(0.331)	(0.635)	(1.897)
Upgraded technique	-0.953	1.639	6.812	-0.457	1.296	5.759
	(0.372)	(0.748)	(2.157)	(0.384)	(0.735)	(2.130)
Process improvement	0.663	2.028	3.016	0.846	1.690	1.870
	(0.318)	(0.664)	(1.947)	(0.330)	(0.652)	(1.930)
Decrease production cost	2.642	1.907	3.971	2.834	1.558	3.175
	(0.287)	(0.585)	(1.856)	(0.297)	(0.571)	(1.833)
Delivery improvement	2.935	2.457	4.918	3.113	2.315	4.654
	(0.286)	(0.574)	(1.821)	(0.296)	(0.565)	(1.809)
R&D Activities						
Basic R&D	1.117	3.177	4.804	0.898	2.291	2.907
	(0.173)	(0.412)	(1.398)	(0.176)	(0.382)	(1.233)
Applied R&D	0.982	2.876	3.611	0.823	2.041	1.726
	(0.171)	(0.387)	(1.273)	(0.177)	(0.365)	(1.181)

Note: Each cell of the table reports the coefficient from a separate linear regression of the innovation activities dependent variable on Only, Diverse Native-, and Diverse Within-Immigrant. Robust standard errors are in parentheses. The first ("+ Finance") specification adds start-up capital amount categories to the base specification, which includes firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience), and the second ("+ Industry") additionally includes 4-digit industry. Coefficients and standard errors are multiplied by 100 for ease of reading. The numbers are weighted by ASE sampling weights. Observations = 199,000 firms. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004; CBDRB-FY23-CES019-010).

Table A.4: Immigrant-Native Diversity, Intellectual Property, Employment, and Productivity

	(1) + Finance			(2) + Industry		
Dependent Variables	Only Immigrant	Diverse Native-Immigrant	Diverse Within-Immigrant	Only Immigrant	Diverse Native-Immigrant	Diverse Within-Immigrant
Intellectual Property						
Copyright or trademark	-0.631	3.301	1.683	-1.300	2.161	1.390
	(0.205)	(0.527)	(1.426)	(0.212)	(0.503)	(1.359)
Patent pending	0.009	1.089	3.040	-0.028	0.815	2.350
	(0.069)	(0.228)	(0.937)	(0.072)	(0.217)	(0.866)
Patent owned	0.124	1.093	2.152	0.059	0.756	1.518
	(0.081)	(0.245)	(0.891)	(0.084)	(0.232)	(0.836)
Patent>0 2012-14	0.124	0.245	1.368	0.116	0.121	1.019
	(0.049)	(0.131)	(0.644)	(0.050)	(0.126)	(0.600)
Patent>2 2012-14	0.043	0.137	0.836	0.034	0.080	0.706
	(0.028)	(0.076)	(0.423)	(0.029)	(0.070)	(0.385)
Patent>0 2015-17	0.084	0.530	1.819	0.079	0.418	1.514
	(0.052)	(0.153)	(0.712)	(0.053)	(0.148)	(0.680)
Patent>2 2015-17	0.064	0.094	D	0.067	0.042	D
	(0.033)	(0.074)	D	(0.034)	(0.073)	D
Employment & Productivity						
Log employment	-0.140	-0.005	0.053	-0.177	-0.019	0.046
	(0.009)	(0.019)	(0.049)	(0.010)	(0.018)	(0.052)
Log productivity	0.098	0.079	-0.032	0.114	0.081	-0.032
	(0.010)	(0.018)	(0.051)	(0.010)	(0.018)	(0.052)
Employment top 5%	-1.437	-0.528	0.189	-1.552	-0.552	-0.182
	(0.153)	(0.434)	(1.148)	(0.162)	(0.417)	(1.177)
Productivity top 5%	1.968	1.214	-0.942	2.141	1.126	-0.982
	(0.250)	(0.473)	(1.302)	(0.260)	(0.471)	(1.294)

Note: Each cell of the table reports the coefficient from a separate linear regression of the intellectual property, employment, and productivity dependent variable on Only, Diverse Native-, and Diverse Within-Immigrant owner. Robust standard errors are in parentheses. The first ("+ Finance") specification adds start-up capital amount categories to the base specification, which includes firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience), and the second ("+ Industry") additionally includes 4-digit industry. Coefficients and standard errors are multiplied by 100 for ease of reading. The numbers are weighted by ASE sampling weights. Observations = 199,000 firms for Intellectual Property and 135,000 for Employment and Productivity. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004; CBDRB-FY23-CES019-010). The cells with "D" indicate the numbers cannot be disclosed due to the limited number of observations.

Table A.5: Low-Tech Immigrant-Native Diversity and Innovation Activities

Dependent Variables	(1) Base		(2) + Finance		(3) + Industry	
	Only Immigrant	Diverse Immigrant	Only Immigrant	Diverse Immigrant	Only Immigrant	Diverse Immigrant
Completely new product	1.994	2.335	1.834	2.219	1.400	1.775
	(0.196)	(0.429)	(0.196)	(0.428)	(0.203)	(0.421)
New product for firm	-2.156	0.186	-2.983	-0.339	-4.328	-1.120
	(0.276)	(0.612)	(0.278)	(0.613)	(0.288)	(0.589)
Other product innovation	5.184	3.364	3.627	2.402	3.345	1.661
	(0.397)	(0.788)	(0.397)	(0.784)	(0.410)	(0.772)
Process innovation	2.971	2.434	1.150	1.337	1.778	1.207
	(0.402)	(0.797)	(0.401)	(0.794)	(0.415)	(0.784)
Basic R&D	0.977	2.204	0.893	2.125	1.031	1.823
	(0.158)	(0.371)	(0.160)	(0.371)	(0.166)	(0.361)
Applied R&D	0.834	1.915	0.780	1.875	0.813	1.531
	(0.160)	(0.350)	(0.160)	(0.350)	(0.168)	(0.343)

Note: Each cell of the table reports the coefficient for a separate LPM regression of the innovation activity and intellectual property dependent variable on Only and Diverse Immigrant owner in low-tech sector. Robust standard errors are in parentheses. The base specification includes firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). The second ("+ Finance") specification adds start-up capital amount categories, and the third ("+ Industry") additionally includes 4-digit industry. Coefficients and standard errors are multiplied by 100 for ease of reading. The numbers are weighted by ASE sampling weights. Observations = 190,000 firms. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004).

Table A.6: Low-Tech Immigrant-Native Diversity, Intellectual Property, Employment, and Productivity

	(1) Base		(2) + Finance		(3) + Industry	
Dependent Variables	Only Immigrant	Diverse Immigrant	Only Immigrant	Diverse Immigrant	Only Immigrant	Diverse Immigrant
Intellectual Property						
Copyright or trademark	-0.299	3.138	-0.540	2.896	-0.891	2.065
	(0.204)	(0.524)	(0.204)	(0.523)	(0.211)	(0.503)
Patent pending	0.006	0.618	-0.020	0.588	-0.003	0.512
	(0.060)	(0.192)	(0.060)	(0.192)	(0.063)	(0.189)
Patent owned	0.144	0.707	0.120	0.675	0.115	0.561
	(0.076)	(0.216)	(0.077)	(0.216)	(0.080)	(0.211)
Patent>0 2012-14	0.092	0.052	0.087	0.041	0.099	0.013
	(0.041)	(0.103)	(0.042)	(0.103)	(0.044)	(0.102)
Patent>2 2012-14	0.024	-0.007	0.023	-0.010	0.022	-0.021
	(0.020)	(0.048)	(0.020)	(0.048)	(0.022)	(0.046)
Patent>0 2015-17	0.068	0.132	0.070	0.124	0.079	0.106
	(0.043)	(0.112)	(0.043)	(0.112)	(0.045)	(0.112)
Patent>2 2015-17	0.056	-0.009	0.059	-0.010	0.065	-0.020
	(0.025)	(0.053)	(0.026)	(0.052)	(0.028)	(0.053)
Employment & Productivity						
Log employment	-0.096	0.061	-0.150	0.035	-0.194	0.015
	(0.010)	(0.021)	(0.010)	(0.020)	(0.010)	(0.019)
Log productivity	0.108	0.093	0.096	0.088	0.112	0.090
	(0.010)	(0.018)	(0.010)	(0.018)	(0.010)	(0.018)
Employment top 5%	-1.533	0.045	-1.674	-0.102	-1.766	-0.179
	(0.155)	(0.456)	(0.156)	(0.452)	(0.166)	(0.433)
Productivity top 5%	1.904	1.216	1.858	1.200	2.013	1.050
	(0.255)	(0.492)	(0.256)	(0.492)	(0.267)	(0.489)

Note: Each cell of the table reports the coefficient for a separate LPM regression of the intellectual property dependent variable on Only and Diverse Immigrant owner in the low-tech sector. Robust standard errors are in parentheses. The base specification includes firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). The second ("+ Finance") specification adds start-up capital amount categories, and the third ("+ Industry") additionally includes 4-digit industry. Coefficients and standard errors are multiplied by 100 for ease of reading. The numbers are weighted by ASE sampling weights. Observations for intellectual property sample = 190,000 firms, and for employment and productivity sample = 129,000 firms. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004).

Table A.7: High-Tech Immigrant-Native Diversity and Innovation Activities

Dependent Variables	(1) Base		(2) + Finance		(3) + Industry	
	Only Immigrant	Diverse Immigrant	Only Immigrant	Diverse Immigrant	Only Immigrant	Diverse Immigrant
Completely new product	-0.124	3.495	-0.351	3.167	-1.429	1.613
	(1.111)	(2.140)	(1.100)	(2.129)	(1.124)	(2.074)
New product for firm	-2.938	4.348	-3.168	3.963	-5.759	2.281
	(1.418)	(2.579)	(1.410)	(2.577)	(1.442)	(2.476)
Other product innovation	3.349	4.265	2.608	3.618	-1.520	1.576
	(1.700)	(2.706)	(1.683)	(2.670)	(1.697)	(2.553)
Process innovation	0.646	7.255	0.205	7.032	-2.117	5.867
	(1.698)	(2.594)	(1.679)	(2.560)	(1.692)	(2.514)
Basic R&D	0.366	11.810	0.010	11.400	-1.141	8.151
	(1.362)	(2.523)	(1.346)	(2.502)	(1.330)	(2.329)
Applied R&D	0.410	10.790	0.116	10.480	0.229	7.755
	(1.303)	(2.460)	(1.291)	(2.452)	(1.274)	(2.309)

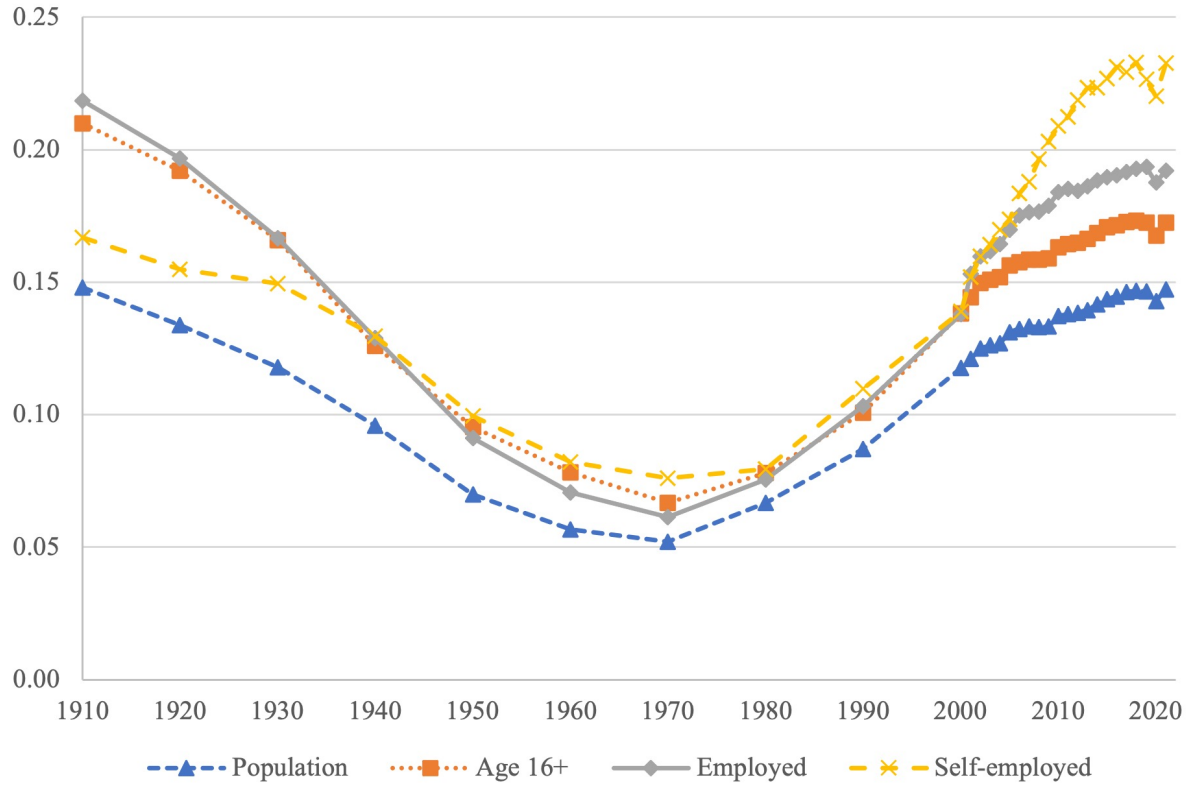
Note: Each cell of the table reports the coefficient for a separate LPM regression of the innovation activity dependent variable on Only and Diverse Immigrant owner in high-tech sector. Robust standard errors are in parentheses. The base specification includes firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). The second ("+ Finance") specification adds start-up capital amount categories, and the third ("+ Industry") additionally includes 4-digit industry. Coefficients and standard errors are multiplied by 100 for ease of reading. The numbers are weighted by ASE regression sampling weights. Observations = 8,900 firms. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004).

Table A.8: High-Tech Immigrant-Native Diversity, Intellectual Property, Employment, and Productivity

Dependent Variables	(1) Base		(2) + Finance		(3) + Industry	
	Only Immigrant	Diverse Immigrant	Only Immigrant	Diverse Immigrant	Only Immigrant	Diverse Immigrant
Intellectual Property						
Copyright or trademark	-5.373	6.980	-5.532	6.784	-7.297	5.560
	(1.212)	(2.565)	(1.207)	(2.566)	(1.228)	(2.492)
Patent pending	0.112	6.184	-0.041	6.000	-0.009	4.633
	(0.704)	(1.696)	(0.685)	(1.685)	(0.682)	(1.598)
Patent owned	-0.261	5.350	-0.266	5.354	-0.449	3.595
	(0.656)	(1.732)	(0.646)	(1.741)	(0.644)	(1.621)
Patent>0 2012-14	0.777	2.457	0.808	2.422	0.781	1.746
	(0.480)	(1.189)	(0.479)	(1.187)	(0.480)	(1.133)
Patent>2 2012-14	0.534	1.840	0.562	1.872	0.553	1.510
	(0.318)	(0.839)	(0.321)	(0.845)	(0.327)	(0.786)
Patent>0 2015-17	0.484	5.189	0.480	5.194	0.533	4.599
	(0.477)	(1.439)	(0.459)	(1.437)	(0.462)	(1.395)
Patent>2 2015-17	0.310	1.210	0.312	1.254	0.371	0.956
	(0.311)	(0.796)	(0.300)	(0.786)	(0.312)	(0.782)
Employment & Productivity						
Log employment	-0.026	0.065	0.001	0.083	0.012	0.064
	(0.040)	(0.070)	(0.039)	(0.068)	(0.039)	(0.067)
Log productivity	0.085	0.057	0.088	0.065	0.093	0.063
	(0.042)	(0.072)	(0.041)	(0.071)	(0.042)	(0.070)
Employment top 5%	0.472	1.474	0.805	1.765	0.347	1.410
	(0.693)	(1.606)	(0.689)	(1.603)	(0.695)	(1.598)
Productivity top 5%	3.339	2.316	3.328	2.231	3.289	2.336
	(1.017)	(1.663)	(1.008)	(1.646)	(1.008)	(1.628)

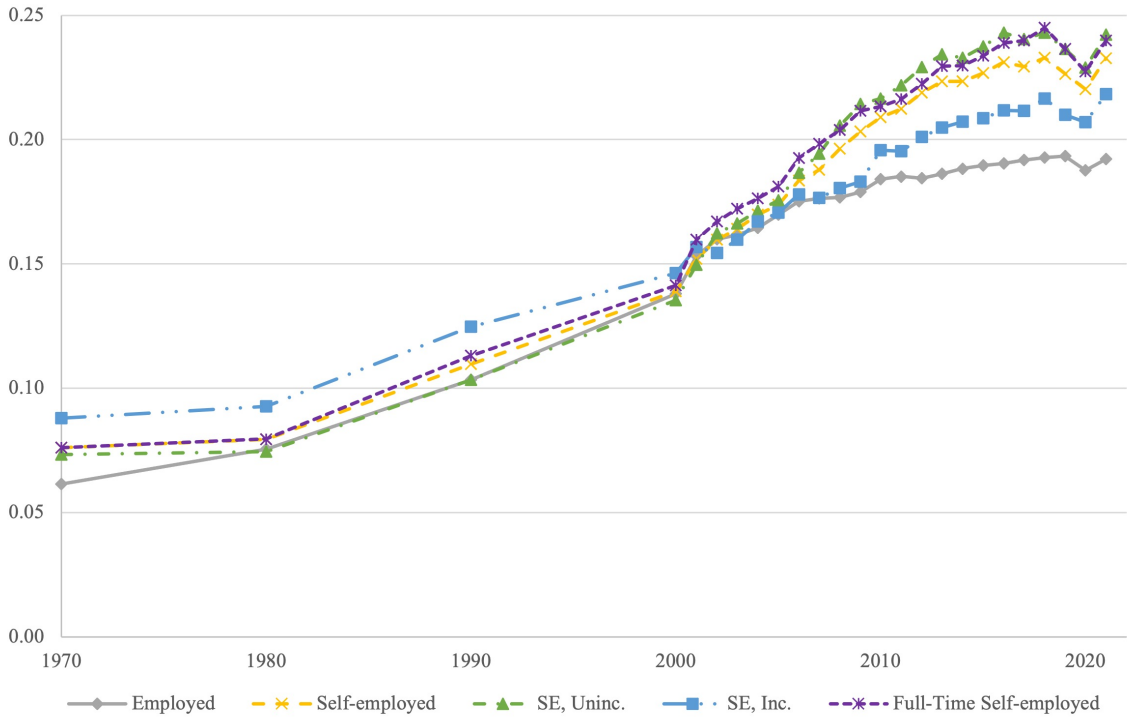
Note: Each cell of the table reports the coefficient for a separate LPM regression of the intellectual property dependent variable on Only and Diverse Immigrant owner in high-tech sector. Robust standard errors are in parentheses. The base specification includes firm age, demographic variables (gender, age, and race/ethnicity), ownership team variables (size and family relationships), and human capital (education, veteran, and prior business experience). The second ("+ Finance") specification adds start-up capital amount categories, and the third ("+ Industry") additionally includes 4-digit industry. Coefficients and standard errors are multiplied by 100 for ease of reading. The numbers are weighted by ASE sampling weights. Observations for intellectual property sample = 8,900 firms, and for employment and productivity sample = 6,700 firms. These results are approved for dissemination by the DRB (CBDRB-FY23-CES019-004).

Figure A.1: Share of Foreign Born by Population Sub-Group, 1910-2021



Note: Data sources include the 1910-2000 Census and 2001-2021 American Community Survey. Estimates are weighted by final person weights provided by the Census Bureau. Foreign born is defined as persons born outside the U.S. or U.S. outlying areas regardless of their citizenship status at birth.

Figure A.2: Share of Foreign Born by Employment and Type of Self-Employment, 1970-2021



Note: Data sources include the 1970-2000 Census and 2001-2021 American Community Survey. Estimates are weighted by final person weights provided by the Census Bureau. Foreign born is defined as persons born outside the U.S. or U.S. outlying areas regardless of their citizenship status at birth.